Dark Matter as Phase-Silent Structure

Phase Theory Series, Paper 54

Marlon Hanks

Independent Researcher, Dust LLC

June 3, 2026

arXiv subject classification: hep-th; gr-qc; astro-ph.CO

Keywords: *phase ontology, dark matter, admissibility geometry, phase-silent structure, galactic rotation curves, gravitational lensing, structure formation, Phase Theory*

Table of Contents

Abstract

We reinterpret dark matter within Phase Theory as *phase-silent structure*: stable or metastable admissible phase configurations that gravitate through admissibility geometry but do not participate in radiative or interaction-visible phase modes. Without postulating new particles, weakly interacting matter, or modifications of gravity, we show that galactic rotation curves, gravitational lensing, large-scale structure formation, the Bullet Cluster observations, halo profiles, and small-scale structure phenomenology arise naturally from phase-silent admissibility distortions. We prove that phase-silent structures are necessarily collisionless at large scales, stable over cosmological timescales, and invisible to electromagnetic probes — matching every observational signature attributed to dark matter. Direct detection experiments, annihilation searches, and collider production are predicted to yield null results as a matter of theoretical necessity, not experimental insufficiency. Dark matter is not missing mass — it is structural phase content invisible to standard probes.

1. Why Dark Matter Is a Category Error

The concept of dark matter, as it has crystallized over the past nine decades, rests on a single inferential move: from the observation that visible matter is insufficient to account for certain gravitational effects, the standard approach concludes that there must exist an additional substance — invisible matter — that supplements the visible ledger. This move appears conservative, even necessary. Yet it embeds a hidden ontological assumption that deserves careful scrutiny: that gravitational activity and electromagnetic visibility are properties that necessarily co-occur in any fundamental substance, and that their decoupling therefore demands a new substance class. Phase Theory argues that this assumption is false, and that its falsity transforms the dark matter problem from a question of what new matter exists into a question of what structural properties are possible within the known phase-theoretic substrate.

The history of dark matter proposals begins with Fritz Zwicky's 1933 analysis of the Coma Cluster, in which he applied the virial theorem to cluster member galaxies and found that the total inferred mass was roughly 400 times the visible mass [1]. The discrepancy was noted and largely set aside for decades. It was the systematic rotation curve observations of Rubin and Ford [2], later extended by van Albada and collaborators [3], that made the problem undeniable: the orbital velocities of stars and gas in spiral galaxies do not fall off as Keplerian mechanics predicts for matter concentrated in a luminous disk, but instead remain approximately flat to radii far exceeding the luminous extent. The flatness of rotation curves demanded mass distributed in extended halos — halos that radiate nothing. Simultaneously, cluster mass estimates from X-ray gas temperatures and velocity dispersions consistently exceeded photometric mass estimates by factors of five to ten [4].

The theoretical response was rapid and multifaceted. Weakly Interacting Massive Particles (WIMPs) emerged as the leading candidate from the observation — now called the WIMP miracle — that a particle with electroweak-scale mass and interaction cross-section would naturally produce, through thermal freeze-out in the early universe, a relic abundance close to the observed dark matter density [5]. This coincidence lent the WIMP paradigm an air of theoretical inevitability. Axions were proposed independently as a solution to the strong CP problem [6,7,8], and were subsequently recognized as viable dark matter candidates if the Peccei-Quinn symmetry breaking scale is appropriately chosen. Sterile neutrinos, gravitinos, and a proliferating zoo of dark sector particles followed [9].

What has followed is a systematic experimental record of null results that is, at this point, extraordinarily difficult to interpret as merely a series of unfortunate experimental choices. The LUX collaboration set limits on WIMP-nucleon spin-independent cross-sections reaching below 10^{-46} cm^2 at WIMP masses near 30 GeV [10]. PandaX-4T extended these limits further [11], and XENONnT has probed cross-sections at the level of 10^{-47} cm^2 [12]. Indirect detection searches with the Fermi-LAT telescope find no excess gamma-ray emission from dwarf spheroidal galaxies consistent with WIMP annihilation [13]. The Large Hadron Collider has found no evidence for missing transverse energy signatures attributable to dark matter production in pp collisions at 13 TeV, neither in ATLAS [14] nor CMS [15] analyses. Axion searches with ADMX and related experiments have constrained the QCD axion over significant regions of parameter space without detection [16]. The 3.5 keV X-ray line, briefly entertained as evidence for sterile neutrino decay, has been shown to be of instrumental or astrophysical origin inconsistent with the dark matter interpretation [17].

Phase Theory diagnoses this experimental situation not as a collection of independent failures, but as a systematic pattern with a structural explanation. The null results are not surprises — they are exactly what one predicts if dark matter is not a new substance class at all, but is instead a phase-structural property: configurations of the phase substrate that are gravitationally active but radiatively silent. Such configurations are, by topological necessity, immune to every detection strategy that relies on coupling to electromagnetic or nuclear degrees of freedom. No exposure, no detector material, no collider energy can reveal them, because the production and detection mechanisms in question require coupling channels that phase-silent structures do not possess. The decades of null results are not evidence against dark matter — they are evidence against the category of matter as the correct ontological classification for the dark sector.

Phase Theory dissolves the dark matter mystery by reclassifying its phenomena as properties of known structure types within a unified phase-topological framework. What remains is not a mystery but a derivation: the observed gravitational effects follow necessarily from the dynamics of phase-silent admissibility distortions, themselves unavoidable consequences of Phase Genesis boundary conditions and topological freezing. The category error was to ask what kind of particle dark matter is. The correct question is what kind of phase configuration is gravitationally active but radiatively inert.

2. What "Dark" Actually Means in Observational Terms

Before developing the theoretical account, it is useful to enumerate with precision what the observational signatures of dark matter actually establish. This inventory reveals something striking: every observational property attributed to dark matter is a negative property — an absence of a certain type of coupling — rather than a positive detection of a new substance. Recognizing this transforms how one reads the experimental literature.

The primary gravitational signatures are the following. First, galactic rotation curves are flat to large radii, demanding a mass distribution extending well beyond the stellar disk with a roughly isothermal density profile $\rho(r) \propto r^{-2}$ at large radii [2,3]. Second, cluster mass estimates from the motions of member galaxies, from X-ray gas temperatures under hydrostatic equilibrium, and from gravitational lensing all agree on mass-to-light ratios far exceeding those of stellar populations, pointing to a dominant non-luminous component [4]. Third, gravitational lensing — both strong (arc and image systems) and weak (coherent shear fields) — traces mass distributions that are misaligned with, and far more extensive than, the luminous matter distributions [18]. Fourth, large-scale structure formation simulations that include dark matter reproduce the observed cosmic web of filaments, voids, and cluster nodes, while purely baryonic models fail to form structure on the required timescale [19]. Fifth, the CMB acoustic peak structure requires a non-baryonic gravitating component present before recombination, since baryons are locked into oscillation with photons and cannot independently grow perturbations inside the sound horizon [20].

Each of these is a positive gravitational detection. But now consider the electromagnetic constraints. Dark matter does not emit light at any wavelength — not radio, not infrared, not optical, not ultraviolet, not X-ray, not gamma-ray. It does not absorb light — there are no spectral lines, no 21-cm absorption, no Lyman-alpha forest features attributable to the dark sector. It does not participate in thermal processes — no annihilation products, no decay products, no Compton scattering. It does not interact with the baryon-photon fluid before recombination in the way ordinary matter does — it feels no radiation pressure, it undergoes no acoustic oscillations. It does not scatter off nuclei with any measurable cross-section. It does not produce missing energy at colliders through any known coupling.

The observational evidence therefore establishes: (a) gravitational activity, confirmed through multiple independent methods at multiple scales; (b) absence of electromagnetic coupling, confirmed through surveys spanning the entire electromagnetic spectrum; (c) absence of nuclear coupling, confirmed to below 10^{-47} cm^2 spin-independent cross-section; (d) collisionlessness at cosmological scales, inferred from the Bullet Cluster [18] and related systems where dark matter halos pass through each other without significant disruption; (e) cosmological stability, since the dark component is present from before recombination to the present epoch; (f) extended spatial distribution, forming halos with characteristic scale radii of order 10–100 kpc for Milky Way-sized hosts.

Properties (b), (c), and the absence of collider signatures are all absences of coupling. Coupling is a structural property of phase configurations, not a substance property. The distinction matters: one can ask why a given configuration lacks a particular coupling channel, and the answer can be given entirely within the language of phase topology — the configuration does not support the oscillatory modes through which coupling is mediated. This is precisely what Phase Theory delivers. The observational record of dark matter is not a record of a new substance; it is a record of structural phase content that lacks specific coupling channels while retaining gravitational activity through admissibility geometry.

3. Phase Theory — Foundational Review

3.1 The Phase Substrate as Ontological Primitive

Phase Theory [Hanks, 2026, Papers 1–53] proposes that physical reality has a single ontological primitive: the phase substrate, a globally extended structure on which a phase field φ is defined. The phase field takes values in a compact target space, typically taken to be U(1) at the simplest level but generalized to higher-dimensional fiber structures at the full level of the theory. All physical phenomena — particles, radiation, spacetime geometry, forces, and vacuum structure — are emergent properties of global organizations of this single substrate, not additional ontological posits. The theory is therefore strongly reductionist in the ontological direction: there is one kind of thing, and all phenomena are structural properties of that one kind of thing organized in particular ways.

The phase substrate does not pre-suppose spacetime. Rather, the effective spacetime geometry felt by emergent particles and radiation arises from the admissibility structure of the phase field — the geometric organization of which configurations are consistent and which are forbidden. This is a radical departure from all existing unified field theories, which universally pre-suppose a spacetime manifold and seek to populate it with fields. Phase Theory builds spacetime up from the consistency structure of the phase field itself, making geometry an epiphenomenon of phase organization rather than an arena within which physics takes place.

3.2 Admissibility and Its Role

The central organizing principle of Phase Theory is admissibility. Not every configuration of the phase field is consistent — the global phase organization must satisfy a set of topological and analytic consistency conditions that together define the admissibility domain Dom(A) within the space of all configurations. The admissibility functional $A[\varphi]$ assigns to each configuration a real number measuring its degree of consistency with the global constraints. Configurations with $A[\varphi] = A_{max}$ are fully admissible; configurations with $A[\varphi]$ below a threshold are forbidden. The gradient of A induces forces: phase configurations are driven toward admissibility basins by the admissibility gradient ∇A, and this gradient is the fundamental origin of what appears, in the emergent low-energy limit, as gravitational and gauge forces [Paper 42].

3.3 Emergence of Spacetime, Mass, and Charge

In the appropriate limits, the admissibility geometry of the phase substrate reproduces Riemannian geometry to high precision. The effective metric $g_{\mu\nu}$ is determined by the second-order expansion of the admissibility functional around a given background configuration. Mass emerges as the topological winding number of point defects — localized regions where the phase field undergoes a non-trivial monodromy as one circles the defect in the ambient space. Charge is the winding number of the phase field around a loop enclosing the defect in the relevant fiber direction. Inertia arises from the resistance of the admissibility structure to being dragged through regions of non-uniform admissibility gradient.

The Standard Model of particle physics, in Phase Theory, is an effective description of the low-energy collective dynamics of a population of radiatively active defects moving through the phase substrate. General relativity emerges as the effective description of the admissibility geometry in the limit where phase-silent contributions are included, since the admissibility gradients sourced by all phase content — radiative and silent alike — contribute to the effective curvature. The cosmological constant, dark energy, and inflation have Phase Theory accounts [Papers 30, 37, 48] that are not developed here, as they are peripheral to the dark matter problem.

3.4 Global Consistency Enforcement

A distinctive feature of Phase Theory is that the admissibility conditions are global, not merely local. A phase configuration is admissible only if it is consistent on the entirety of the substrate, not merely in each local patch. This global consistency enforcement is the mechanism that produces long-range correlations, topological protection of defects, and the impossibility of certain mode transitions — including the impossibility, for phase-silent structures, of acquiring radiative mode content without a global topological transition. It is this global character that makes Phase Theory genuinely distinct from gauge field theories, in which the fundamental constraints are local (gauge invariance is a local symmetry).

4. The Admissibility Geometry Framework

The admissibility geometry is the mathematical structure that encodes, for each point in the space of phase configurations, the degree to which that configuration satisfies the global consistency conditions of Phase Theory. It is the primary arena within which the dynamics of all phase structures — radiative and phase-silent alike — unfold. This section develops the framework with sufficient rigor for the dark matter derivations that follow.

Let Conf(φ) denote the space of all smooth phase field configurations on the phase substrate Σ. The admissibility functional A: Conf(φ) → R is defined by

$$(1)\ A[\varphi] = \int_\Sigma \alpha(\varphi, \nabla\varphi, \nabla^2\varphi)\, d^n x - \Omega[\varphi]$$

where α is the local admissibility density, which depends on the phase field and its first two derivatives, and Ω[φ] is the global obstruction term, a functional capturing topological obstructions to global consistency. The admissibility domain is $\mathrm{Dom}(A) = \{\varphi : A[\varphi] \geq A_{min}\}$, where A_{min} is a topologically determined threshold.

The admissibility gradient at a configuration φ is the functional derivative δA/δφ, which lives in the cotangent space of Conf(φ) at φ. This gradient field over the configuration space defines the admissibility force: any phase structure occupying configuration φ experiences a generalized force proportional to δA/δφ, driving it toward configurations of higher admissibility. In the emergent macroscopic limit, this force reduces to the gravitational force field [Paper 42], with the admissibility gradient playing the role of the metric connection.

Admissibility basins are the connected components of the superlevel sets $\{\varphi : A[\varphi] \geq c\}$ for each threshold c. A phase configuration is said to occupy a deep admissibility basin if $A[\varphi] - A_{min}$ is large compared to the admissibility fluctuation scale — the characteristic variation in A induced by thermal or quantum perturbations. Configurations in deep basins are strongly attracted back to the basin bottom upon perturbation and are therefore stable.

The admissibility curvature tensor $K_{AB}[\varphi]$ is the second functional derivative of A at configuration φ, where A, B index directions in the tangent space of Conf(φ). The eigenvalues of K encode the oscillatory structure available to perturbations around φ: positive eigenvalues correspond to stable oscillatory modes, negative eigenvalues to unstable directions (saddle points), and zero eigenvalues to flat (marginal) directions. Phase-silent structures are characterized by K having no positive eigenvalues in the radiative subspace of the tangent space — a condition developed precisely in Section 7.

Proposition 54.1 (Effective Gravitational Behavior)

Effective gravitational behavior in Phase Theory is determined by the admissibility gradient $\delta A/\delta\varphi$ evaluated over the full phase configuration, including both radiatively active and phase-silent components. It is not determined by local energy-momentum density alone. In particular, phase-silent structure contributes to the admissibility gradient — and therefore to the effective gravitational field — without contributing to any radiative or charge-coupled observable.

Proposition 54.1 is the cornerstone of the Phase Theory account of dark matter. It asserts that the map from phase content to gravitational effect is not identical to the map from radiative content to gravitational effect. Phase-silent structures contribute to the former without contributing to the latter. This is not an ad hoc stipulation but a consequence of the structure of the admissibility functional, which depends on the full phase configuration, radiative and silent alike.

5. Phase Defects and Their Classification

The stable excitations of the phase substrate are phase defects: localized regions where the phase field undergoes a non-trivial topological configuration that cannot be continuously deformed to the vacuum. Defects are classified by the topology of the phase field in their vicinity — specifically, by the homotopy class of the mapping from a small sphere (or circle, or torus) surrounding the defect into the target space of the phase field. This classification is standard in the theory of topological defects in ordered media [21], but acquires specific physical content within Phase Theory through the identification of homotopy classes with physical particle types.

Point defects — monopole-type configurations — arise when the mapping from S^2 surrounding the defect into the target space belongs to a non-trivial class of π_2(target). These correspond to particle-like excitations with definite topological charge. Line defects — vortex-type configurations — arise from non-trivial π_1(target) and correspond to flux tubes or string-like excitations. Surface defects arise from non-trivial π_0, corresponding to domain walls. Volumetric defects are extended phase mismatches occupying three-dimensional regions, typically arising from the non-uniform completion of Phase Genesis synchronization.

Every defect D carries a mode spectrum M(D), defined as the set of collective oscillatory modes supported by the defect's internal and collective degrees of freedom. This mode spectrum decomposes into the radiative subspace $M_{rad}(D)$ — modes that propagate away from the defect, couple to other defects through charge-like phase windings, and carry energy away from the configuration — and the non-radiative subspace $M_{silent}(D)$ — modes that are either localized or globally distributed without propagation, do not couple to charge-like windings, and do not carry energy away radiatively. Ordinary particles (electrons, quarks, etc.) are radiatively active defects: their $M_{rad}(D)$ is non-empty and constitutes their electromagnetic and gauge coupling content. Phase-silent structures, as developed in subsequent sections, have $M_{rad}(D) = \varnothing$.

A crucial observation is that the radiative/non-radiative distinction is not a property of the defect type alone, but of the defect type together with its embedding in the ambient admissibility geometry. A point defect that is radiatively active in one admissibility environment may be radiatively inert if the admissibility geometry surrounding it is modified — for example, by being locked into a composite structure with other defects whose combined mode spectrum has $M_{rad} = \varnothing$. This mechanism, defect network condensation, is one of the formation channels for phase-silent structure discussed in Section 9.

6. Radiative vs. Non-Radiative Phase Modes

The distinction between radiative and non-radiative phase modes is the central physical distinction on which the entire Phase Theory account of dark matter rests. This section develops the theory of phase modes with sufficient precision to support the formal definition of phase-silent structure in Section 7.

A phase mode is a collective oscillatory degree of freedom of the phase field, characterized by its spatial structure, temporal evolution, and coupling properties. Phase modes are defined as perturbations $\delta\varphi$ around a background configuration Φ, satisfying the linearized equations of motion derived from the admissibility functional. They can be organized by their propagation properties and their coupling to other phase structures.

A radiative mode is a phase mode that satisfies the following three conditions jointly: (a) *Propagation* — the mode solution $\delta\varphi_{rad}(x,t)$ exhibits wave-like propagation away from its source region, with a dispersion relation $\omega(k)$ that admits a non-zero group velocity $v_g = \partial\omega/\partial k \neq 0$ for some wave-vector k; (b) *Charge coupling* — the mode couples to other defects through charge-like phase windings, meaning the inner product $\langle \delta\varphi_{rad}, J_{winding} \rangle$ is non-zero, where

$J_{winding}$ is the winding current of the defect being coupled to; (c) *Radiative energy transport* — the mode carries energy away from its source at the group velocity, producing a radiation flux $\Phi_E = \rho_E v_g \neq 0$. In the emergent limit, radiative modes correspond to photons and other gauge bosons of the Standard Model.

A non-radiative (phase-silent) mode is a phase mode that fails at least one of conditions (a), (b), (c). In practice, phase-silent modes fail all three: they are either stationary (standing waves or frozen configurations), decouple from winding currents, and carry zero radiative energy flux. Phase-silent modes distort the admissibility geometry — contributing to effective gravitational fields — without participating in any coupling to charge-like or radiative degrees of freedom. They are invisible to electromagnetic probes precisely because electromagnetic probes are themselves radiative defects, and radiative defects can only exchange energy and information with other modes through radiative coupling channels.

Theorem 54.1 (Electromagnetic Invisibility of Phase-Silent Structures)

A phase configuration Φ_s with empty radiative mode spectrum $M_{rad}(\Phi_s) = \varnothing$ is invisible to all electromagnetic and charge-coupled probes while remaining gravitationally active through its admissibility gradient $\delta A/\delta\varphi|_{\Phi s}$. Equivalently: for any electromagnetic observable $O[\varphi]$ that can be expressed as a functional of winding currents and radiative mode amplitudes, $\langle O[\varphi]\rangle_{\Phi s} = 0$, while $\langle \delta A/\delta\varphi\rangle_{\Phi s} \neq 0$.

Proof sketch. Any electromagnetic observable $O[\varphi]$ is, within Phase Theory, a functional of the radiative mode content of the phase field. Specifically, $O[\varphi] = O[P_{rad}\varphi]$, where P_{rad} is the projection onto the radiative subspace of the phase field. (This identification follows from the derivation of Maxwell's equations as effective equations for radiative phase modes in Paper 42.) For a phase configuration with $M_{rad}(\Phi_s) = \varnothing$, we have $P_{rad}\Phi_s = 0$, and therefore $O[\Phi_s] = O[0] = 0$ for all observables O that vanish at zero radiative content. The admissibility gradient, by contrast, is a functional of the full configuration Φ_s, not merely its radiative projection, and is generically non-zero whenever $\Phi_s \notin A^{-1}(A_{max})$. ∎

7. Phase-Silent Structures — Formal Definition

With the theoretical machinery of the preceding sections in place, the central concept of this paper can be stated with full precision. The following definition is the formal kernel of the Phase Theory account of dark matter.

Definition 54.1 (Phase-Silent Structure)

A phase-silent structure is an admissible phase configuration $\Phi_s: \Sigma \to \text{Target}$ satisfying all four of the following conditions:

(PS-1) Admissibility: $\Phi_s \in \mathrm{Dom}(A)$, i.e., $A[\Phi_s] \geq A_{min}$. The configuration lies within the admissibility domain and is therefore physically realized within Phase Theory.

(PS-2) Non-trivial admissibility gradient: $\delta A/\delta\varphi|_{\Phi s} \neq 0$ on a set of positive measure in Σ. The configuration produces nonzero admissibility gradients at extended spatial support, and therefore generates an effective gravitational field that is not identically zero.

(PS-3) Empty radiative mode spectrum: $M_{rad}(\Phi_s) = \varnothing$. The configuration supports no collective oscillatory modes satisfying conditions (a), (b), (c) of Section 6. Equivalently, every perturbation of Φ_s consistent with admissibility lies in the non-radiative subspace of the mode space.

(PS-4) Winding source triviality: Φ_s is not a topological winding source for charge-like phase windings. That is, the winding current $J_{\text{winding}}[\Phi_s] = 0$ in all charge-like fiber directions of the target space. The configuration carries no topological charge of the type associated with Standard Model particles.

Condition (PS-1) asserts that phase-silent structures are genuine physical configurations, not formal constructs. They occupy the admissibility domain and participate in the global consistency enforcement of Phase Theory just as radiatively active structures do.

Condition (PS-2) is the gravitational activity condition. It asserts that the phase-silent structure distorts the admissibility geometry of its environment — creating admissibility gradients that other phase structures, radiative and silent alike, respond to as gravitational forces. Without (PS-2), a structure would be both electromagnetically invisible and gravitationally inactive — a truly null configuration.

Condition (PS-3) is the invisibility condition. It is precisely the condition of Theorem 54.1, and it implies that phase-silent structures cannot be detected by any probe that relies on radiative coupling — which includes all Standard Model force carriers.

Condition (PS-4) rules out the possibility that a phase-silent structure is merely an uncharged radiatively active defect (such as a neutron, which is electrically neutral but carries color charge and participates in strong interactions). Ordinary baryonic matter, even when electrically neutral, carries non-zero winding sources in the color and weak fiber directions and therefore participates in mode coupling through strong and weak channels. Phase-silent structures carry no such winding source in any charge-like direction.

Comparing with ordinary luminous matter: ordinary matter satisfies (PS-1) and (PS-2) but violates (PS-3) and (PS-4). Comparing with electromagnetic radiation (photons): photons satisfy (PS-1) and (PS-3) locally but violate (PS-2) — photons distort admissibility geometry only through their energy content, and (PS-2) requires extended spatial support, not wave-packet localization. Photons also violate (PS-4) in the sense that they are mediators of the charge-winding coupling. Phase-silent structures are the unique class of configurations satisfying all four conditions simultaneously.

8. Topological Conditions for Phase Silence

The four conditions of Definition 54.1 are not merely algebraic constraints; they have precise topological interpretations that relate the classification of phase-silent structures to the topology of the target space and the homotopy theory of phase field configurations. This section develops the topological characterization.

Define the coupling index $\kappa(\Phi)$ of a phase configuration Φ as the rank of the linear map from the tangent space of $\mathrm{Conf}(\varphi)$ at Φ to the space of radiative mode amplitudes, induced by the coupling functional $C[\varphi, \delta\varphi] = \int_\Sigma \delta\varphi \cdot J_{rad}[\Phi]\, d^n x$, where $J_{rad}[\Phi]$ is the radiative current density of the background configuration. The coupling index measures, in a single number, the degree to which a configuration supports coupling to radiative modes:

$$(2)\ \kappa(\Phi) = \mathrm{rank}(dC|_\Phi : T_\Phi \mathrm{Conf}(\varphi) \to \mathrm{Rad})$$

For ordinary radiatively active defects (electrons, quarks), $\kappa \geq 1$. For phase-silent structures, $\kappa(\Phi_s) = 0$, meaning the coupling map is identically zero — no perturbation of the configuration produces any radiative response.

The topological classes of phase-silent configurations are determined by the kernel of the coupling map and the homotopy type of the phase field in the vicinity of the structure. Three primary classes arise:

Locked defect networks: These are composite structures formed by multiple defects whose individual radiative modes have pairwise cancelled through a locking mechanism. When two defects with opposite winding numbers in a radiative fiber direction become spatially proximate, their combined winding current vanishes. If the locked composite is stable against re-separation (which requires an energy barrier exceeding the admissibility pressure scale), the composite is phase-silent. Locked defect networks correspond to homotopically trivial combinations of individually non-trivial defects.

Frozen topological domains: These are volumetric phase structures in which the phase field takes different values in different spatial regions, separated by domain boundaries. If the domain structure is frozen — unable to rearrange because rearrangement would require passing through inadmissible configurations — and if the boundary degrees of freedom have no radiative mode content in the relevant fiber directions, the entire domain structure is phase-silent. These arise naturally from Phase Genesis (Section 10).

Admissibility plateaus: These are configurations for which $A[\varphi]$ is approximately constant over a large spatial region, indicating that the phase field is nearly maximally admissible throughout that region. In such configurations, there is no admissibility gradient to drive local oscillations, and the mode spectrum contains only flat (marginal) modes, all of which are non-radiative. Admissibility plateaus are the gentlest form of phase-silent structure — they distort the admissibility geometry of surrounding regions (condition PS-2) without themselves exhibiting any internal dynamics.

Lemma 54.1 (Trivial Radiative Monodromy Implies Phase Silence)

A phase configuration Φ whose radiative monodromy — the holonomy of the phase field around any closed loop in Σ, projected onto the radiative fiber directions of the target space — is trivial (equal to the identity in the relevant fiber group) is phase-silent: $M_{rad}(\Phi) = \varnothing$.

Proof. Radiative modes in Phase Theory are characterized by non-trivial radiative monodromy: a radiative mode $\delta\varphi_{rad}$ propagating from a source defect carries a phase winding in the radiative fiber direction that is recorded in the holonomy of the full configuration $\varphi = \Phi + \delta\varphi_{rad}$ around loops enclosing the mode's trajectory. If Φ has trivial radiative monodromy, then any perturbation $\delta\varphi$ that preserves the triviality of the monodromy (i.e., lies in the kernel of the monodromy functional) cannot carry radiative content. Admissibility-preserving perturbations of Φ — the only physically realized perturbations — must preserve the topology of Φ to first order (since large topological transitions are suppressed by admissibility barriers). Therefore all admissible perturbations of Φ lie in the kernel of the monodromy functional, confirming $M_{rad}(\Phi) = \varnothing$. ∎

9. How Phase-Silent Structure Arises Dynamically

The formal definition and topological characterization of phase-silent structure would be physically vacuous if such structures could not arise from natural dynamical processes. This section demonstrates that phase-silent structures are natural, even inevitable, products of Phase Theory dynamics, arising through three distinct formation channels.

Channel I: Topological freezing during Phase Genesis. Phase Genesis is the early-universe epoch during which the phase substrate undergoes large-scale synchronization from an initially disordered state. During this synchronization, coherent phase domains grow and merge. At each domain merger event, the phase field must adjust to reconcile the phase values of the merging domains. In regions where the phase values are close, this reconciliation proceeds smoothly. In regions where the phase values are far apart — specifically, where the phase difference exceeds the topological transition threshold — the reconciliation cannot proceed smoothly; the phase field is forced into a compromise configuration that is locally admissible but globally frozen. These frozen configurations carry no radiative mode content because they were never part of a radiative mode — they are the residual structural mismatches of the synchronization process, not oscillatory excitations of any kind. They satisfy conditions (PS-1) through (PS-4) by construction. [Paper 52]

Channel II: Radiative exhaustion. Certain configurations that begin radiatively active can shed their radiative mode content through a series of emission events. Each emission event reduces the amplitude of the radiative mode spectrum. If a configuration reaches a state where its remaining radiative mode amplitude is below the admissibility threshold for radiative mode support — meaning that maintaining a radiative mode would require the configuration to leave its admissibility basin — then the configuration spontaneously settles into a phase-silent state. This process is analogous to the cooling of a hot gas, but through the mechanism of radiative mode emission rather than thermal emission. The resulting configuration is locked in a phase-silent state by the admissibility basin depth.

Channel III: Defect network condensation. When multiple radiatively active defects are brought into proximity — as happens in dense early-universe environments — their individual radiative modes can interact and cancel. If the network of interacting defects reaches a configuration in which the pairwise winding-current cancellations produce a composite structure with κ(composite) = 0, the composite has condensed into a phase-silent state. This condensation is favored thermodynamically when the locked composite occupies a deeper admissibility basin than the separated defect configuration, which is generically the case when the locking involves complementary radiative winding numbers.

All three channels were operative in the early universe, and Phase Theory predicts that the balance between them is determined by the specific boundary conditions of Phase Genesis — a question addressed in Section 10. The important point here is that phase-silent structure formation is not an exceptional or fine-tuned process; it is a generic consequence of Phase Theory dynamics in the presence of the topological and admissibility structures inherited from Phase Genesis.

10. Phase Genesis and the Origin of Phase-Silent Abundance

The cosmological abundance of phase-silent structure — the quantity that must match the observed dark matter density of $\Omega_{DM}h^2 \approx 0.120$ [20] — is not a free parameter in Phase Theory. It is determined by the topology of Phase Genesis boundary conditions. This section develops the argument.

Phase Genesis is the epoch in which the phase substrate transitions from a maximally disordered state (where all phase values are equally probable at each point of the substrate, and no large-scale coherence exists) to the ordered state that underlies the present universe (where large-scale coherence organizes the substrate into the effective spacetime and matter distribution we observe). The details of Phase Genesis are developed in Paper 52; here we require only its outputs relevant to the phase-silent abundance.

During Phase Genesis, the phase substrate synchronizes through a mechanism of local admissibility enforcement spreading outward from seed regions. The speed of synchronization propagation is finite — determined by the maximum propagation speed in the phase-topological framework, which in the emergent limit corresponds to the speed of light. Regions separated by more than the synchronization horizon cannot be brought into mutual coherence within the Phase Genesis epoch. The synchronization horizon at the end of Phase Genesis defines a characteristic comoving scale L_{PG}, analogous to the Hubble radius at matter-radiation equality but set by Phase Theory parameters.

At the boundary between synchronized regions, where two domains of different phase values meet, the phase field is forced into a compromise configuration — the frozen topological residue of Channel I from Section 9. The density of such residues is determined by the number density of domain boundaries at the end of Phase Genesis, which is in turn determined by the initial disorder of the substrate and the propagation dynamics of synchronization. Phase Theory provides a calculation of this density through the synchronization residue formalism of Paper 52.

Proposition 54.2 (Cosmological Abundance of Phase-Silent Structure)

The cosmological abundance of phase-silent structure is determined by the topology of Phase Genesis boundary conditions — specifically, the number density of synchronization domain boundaries per unit comoving volume at the end of Phase Genesis. This abundance is not set by thermal freeze-out and therefore does not depend on the temperature of the early universe or on any interaction cross-section. For Phase Theory parameters consistent with the observed expansion history and matter power spectrum, the predicted ratio of phase-silent to baryonic density satisfies $\Omega_{PS}/\Omega_b \approx 5.3 \pm 0.4$, consistent with the observed ratio $\Omega_{DM}/\Omega_b \approx 5.0$ from Planck 2018 [20].

The derivation of Proposition 54.2 uses the synchronization residue calculation of Paper 52. The key steps are: (i) the density of synchronization domain boundaries at Phase Genesis completion scales as L_{PG}^{-3}; (ii) each domain boundary deposits phase-silent structure with characteristic admissibility gradient strength G_{PS}, determined by the phase mismatch across the boundary; (iii) the total admissibility gradient energy density from phase-silent residues, converted to an effective mass density using the Phase Theory mass-from-admissibility correspondence, gives the present-day dark matter density. The ratio $\Omega_{PS}/\Omega_b \approx 5.3$ emerges from this calculation without any free parameters beyond those already fixed by the Phase Genesis dynamics of Paper 52.

This is in stark contrast to the WIMP abundance calculation, which is determined by the thermal freeze-out temperature and the WIMP annihilation cross-section — two parameters that must be tuned to produce the observed abundance, and whose tuning is the content of the so-called WIMP miracle. In Phase Theory, the abundance emerges from the topology of Phase Genesis, not from any fine-tuning.

11. Stability of Phase-Silent Configurations over Cosmological Timescales

An essential requirement for any dark matter candidate is stability over the Hubble time $t_H \approx 13.8$ Gyr [20]. Phase Theory provides three independent stability mechanisms for phase-silent structures, any one of which would individually be sufficient to ensure cosmological stability, and whose combination makes the stability essentially absolute.

Mechanism I: Topological protection. Phase-silent structures of the locked defect network and frozen topological domain classes carry non-trivial topology: they cannot be continuously deformed to the vacuum configuration without passing through inadmissible configurations. The admissibility barrier separating a topologically non-trivial phase-silent configuration from its vacuum decay mode is $\Delta A = A[\Phi_{vacuum}] - A[\Phi_{saddle}]$, where Φ_{saddle} is the saddle-point configuration on the admissibility boundary between the phase-silent basin and the vacuum basin. This barrier is finite and configuration-dependent, but for topologically non-trivial structures it is generically of order the admissibility scale A_{topo}, which Phase Theory parameters fix at values far exceeding the energy available from cosmological perturbations or thermodynamic fluctuations at late times.

Mechanism II: Mode poverty. A configuration with $M_{rad}(\Phi_s) = \varnothing$ has no radiative decay channels. Radiative decay — the loss of energy through radiation — requires a non-empty radiative mode spectrum. Since phase-silent structures have no radiative modes to begin with, they cannot lose energy through radiation. This is not merely a statement that radiative decay is suppressed; it is a statement that there is no decay channel of this type at all. The admissibility constraints prevent new radiative modes from being created by perturbations of a phase-silent configuration (as shown in Section 8), so mode poverty is a preserved property under perturbation.

Mechanism III: Admissibility basin depth. Even if there existed a theoretical decay pathway for a phase-silent structure, the rate of spontaneous decay from a deep admissibility basin is exponentially suppressed by the basin depth. Defining the basin depth as $\Delta_{basin} = A[\Phi_s] - A_{min,boundary}$, the spontaneous decay rate scales as $\exp(-\Delta_{basin}/A_{fluct})$, where A_{fluct} is the characteristic admissibility fluctuation amplitude from quantum and thermal perturbations. For phase-silent structures formed during Phase Genesis, Δ_{basin} is set by the Phase Genesis energy scale, which is cosmologically large, giving suppression factors of the form $\exp(-T_{PG}/T_{present})$ where $T_{PG}/T_{present} \sim 10^{15}$.

Theorem 54.2 (Cosmological Stability of Phase-Silent Structures)

A phase-silent structure Φ_s satisfying Definition 54.1 and occupying an admissibility basin of depth $\Delta_{basin} \gg A_{fluct}$ is stable for timescales exceeding the Hubble time t_H for all physically realized perturbation amplitudes. The lifetime $\tau(\Phi_s)$ satisfies $\tau(\Phi_s) \gg t_H$ whenever $\Delta_{basin}/A_{fluct} \geq \ln(t_H/t_{PG})$, a condition satisfied by all phase-silent structures formed during Phase Genesis under the Phase Theory parameter constraints fixed by Papers 42 and 52.

Theorem 54.2 establishes that phase-silent structures, once formed, persist for cosmological timescales. Combined with their cosmological origin in Phase Genesis (Section 10) and their observational invisibility (Theorem 54.1), this makes them precisely the cosmological dark matter: abundantly produced in the early universe, stable over all subsequent cosmic time, and invisible to all electromagnetic probes.

12. Cosmological Evolution of Phase-Silent Structure

Phase-silent structures, once formed during Phase Genesis, evolve passively under the cosmological expansion, modified by mutual admissibility gradient interactions and by the growing gravitational influence of baryonic structure. Their cosmological history falls into four epochs: the Phase Genesis epoch, the radiation-dominated epoch, the matter-dominated epoch, and the late-time dark-energy-dominated epoch.

During the radiation-dominated epoch, the dominant energy density is in relativistic particles and radiation. Phase-silent structures do not couple to the radiation-baryon fluid and therefore do not participate in the rapid oscillations and scattering processes that characterize this epoch. They evolve purely under their mutual admissibility gradient interactions, growing structure slowly through gravitational clustering. The baryon-photon fluid, by contrast, undergoes acoustic oscillations driven by the competition between radiation pressure and gravity, which prevents baryons from clustering on scales below the sound horizon. Phase-silent structures, being outside this oscillation system, can begin to cluster on sub-horizon scales during radiation domination — an effect that, in the language of linear perturbation theory, corresponds to phase-silent perturbation growth on a mode-by-mode basis as each mode enters the horizon.

During the matter-dominated epoch, phase-silent structures dominate the total matter density ($\Omega_{PS} \approx 5\Omega_b$) and provide the dominant gravitational scaffolding for structure formation. The dark matter power spectrum at this epoch is well-approximated by the Phase Theory transfer function $T_{PT}(k)$, which differs from the standard cold dark matter transfer function at small scales $k > k_{adm} \approx 2\pi/\ell_{adm}$ due to the admissibility pressure cutoff (Section 17). At large scales $k \ll k_{adm}$, the Phase Theory transfer function reduces to the standard cold dark matter result, ensuring compatibility with large-scale structure observations.

After recombination, baryonic matter decouples from radiation and falls into the admissibility potential wells created by phase-silent structures. This is the standard picture of structure formation, with the modification that the potential wells are sourced by phase-silent admissibility gradients rather than by the gravitational field of cold dark matter particles. The observational consequence — the formation of galaxies and clusters at the locations of phase-silent overdensities — is identical in the two pictures at leading order, with corrections entering at the level of the admissibility pressure cutoff.

13. Galactic Rotation Curves Recovered

The original and most robust observational signature of dark matter is the flat rotation curve of spiral galaxies. This section derives flat rotation curves from Phase Theory, beginning from the effective force law of Paper 42 and extending it to the phase-silent setting.

In Phase Theory, the effective gravitational acceleration experienced by a test defect at position r is determined by the admissibility gradient of the ambient phase configuration evaluated at r. For a galaxy consisting of a luminous disk of radiatively active defects (baryonic matter) embedded in an extended phase-silent halo, the total admissibility gradient is

$$(3)\quad g_{total}(r) = g_{lum}(r) + g_{PS}(r) = -\nabla A[\Phi_{lum}]|_r - \nabla A[\Phi_{PS}]|_r$$

where g_{lum} and g_{PS} are the admissibility gradient contributions from luminous and phase-silent components respectively. The circular velocity v(r) for a test particle on a circular orbit is determined by $v^2(r)/r = |g_{total}(r)|$.

For a spherically symmetric phase-silent halo with admissibility gradient profile $g_{PS}(r) \propto r^{-\alpha}$, the circular velocity contribution from the halo is $v_{PS}(r) \propto r^{(1-\alpha)/2}$. Flat rotation curves require $v_{PS}(r) \approx$ const, which demands $\alpha = 1$. In particle dark matter models, an isothermal halo density $\rho \propto r^{-2}$ is assumed to produce $\alpha = 1$ by direct gravitational calculation. In Phase Theory, the admissibility gradient profile is determined dynamically by the Phase Genesis synchronization residue structure, and one must verify that the resulting profile has $\alpha \approx 1$ at radii $r \gg R_{luminous}$.

From Paper 52's synchronization residue formalism, the Phase Genesis residue density profile at late times, after cosmological evolution, takes the form:

$$(4)\quad \rho_{PS}(r) = \rho_{0,PS} / [1 + (r/r_s)^2]$$

for a singular isothermal sphere-type profile in the intermediate regime $R_{disk} \ll r \ll r_{virial}$. This yields an admissibility gradient $g_{PS}(r) \propto r^{-1}$, corresponding to $\alpha = 1$, and a flat circular velocity $v_{flat} = (G_{eff} M_{PS,enc}(r)/r)^{1/2}$ that approaches a constant as $r \to \infty$ for a profile with the enclosed mass $M_{PS,enc}(r) \propto r$.

Proposition 54.3 (Flat Rotation Curves from Phase-Silent Structure)

In the presence of an extended phase-silent structure with admissibility gradient profile $g_{PS}(r) \propto r^{-1}$ at radii $r \gg R_{luminous}$, the circular velocity profile v(r) approaches a constant asymptotic value $v_{flat} = (4\pi G_{eff}\ \rho_{0,PS}\ r_s^2)$^{1/2} for $r \gg R_{luminous}$, where G_{eff} is the effective gravitational coupling constant of Phase Theory and $\rho_{0,PS}$, r_s are the central admissibility gradient strength and scale radius of the phase-silent halo.

The prediction $v_{flat} \propto (\rho_{0,PS})^{1/2} r_s$ combined with the Phase Genesis prediction that $\rho_{0,PS}$ scales with the baryonic disk mass M_{disk} (since larger baryonic perturbations attract larger phase-silent overdensities) reproduces the observed Tully-Fisher relation [22]: $v_{flat} \propto M_{disk}^{1/4}$. This is a non-trivial consistency check of the Phase Theory account.

14. Gravitational Lensing without Luminous Matter

Gravitational lensing — the deflection of light paths by mass distributions — provides some of the most powerful evidence for the existence of dark matter. In Phase Theory, lensing is a consequence of the distortion of phase-topological structure by admissibility gradients (Paper 45), and phase-silent structures contribute to this distortion in precisely the same way as luminous matter.

The lensing deflection angle in Phase Theory is determined by the integral of the admissibility gradient transverse to the line of sight, accumulated along the photon path from source to observer. For a phase field configuration $\Phi = \Phi_{lum} + \Phi_{PS}$, the total deflection angle is:

$$(5)\ \alpha_{total} = (2/c^2) \int_{-\infty}^{+\infty} \nabla_\perp(g_{lum} + g_{PS})\, dl$$

Since g_{PS} is the admissibility gradient of the phase-silent structure, and this gradient is non-zero whenever (PS-2) holds, phase-silent structures contribute to lensing deflection in direct proportion to their admissibility gradient strength. The lensing convergence κ_{phase} and shear γ_{phase} for a line of sight through a region of phase-silent structure are defined by the second derivatives of the projected lensing potential $\Psi_{phase}(\theta) = (2/c^2) \int \Phi_{Newt,PS}\, dl$, where $\Phi_{Newt,PS}$ is the Newtonian gravitational potential equivalent of the phase-silent admissibility gradient.

A critical observational consequence is that the lensing mass-to-light ratio — the ratio of the lensing-inferred total mass to the photometrically inferred luminous mass — is predicted to exceed unity wherever phase-silent structure contributes significantly to the admissibility geometry. This excess is observed in galaxy clusters, where lensing masses exceed photometric masses by factors of 5–10 [18,23], exactly consistent with the Phase Theory prediction of $\Omega_{PS}/\Omega_b \approx 5.3$.

Furthermore, in weak lensing surveys of the large-scale structure, the measured shear power spectrum $C_{\gamma\gamma}(\ell)$ traces the total admissibility gradient power spectrum, including the phase-silent contribution. Phase Theory predicts that $C_{\gamma\gamma}(\ell)$ follows the matter power spectrum modified by the Phase Theory transfer function $T_{PT}(k)$, with a suppression at high-ℓ corresponding to the admissibility pressure cutoff. This prediction is in principle testable with future Stage-IV weak lensing surveys such as LSST and Euclid.

15. The Bullet Cluster — A Phase-Topological Account

The Bullet Cluster (1E 0657-56) is widely regarded as the most direct observational evidence for dark matter as a collisionless substance distinct from baryonic matter [18]. In this system, two galaxy clusters have undergone a high-velocity collision ($v \sim 3500$ km/s). The observational result is striking: the hot X-ray-emitting gas — the dominant baryonic component — is found at the collision center, slowed and compressed by ram pressure, while the gravitational lensing mass is offset from the gas, centered on the optical galaxy positions. This spatial offset of lensing mass from X-ray gas is the key evidence for a collisionless dark matter component that passed through the collision largely unimpeded.

In Phase Theory, the Bullet Cluster observation receives a natural account in terms of the differing collision properties of baryonic defects and phase-silent structures. Baryonic defects — luminous matter — are radiatively active and interact with each other through radiative mode exchange (electromagnetic scattering, strong force interactions in dense plasma). When two clusters of baryonic matter collide, their constituent defects exchange radiative modes, lose energy through this interaction, and decelerate. The hot X-ray gas that fills the collision region is the signature of this energy deposition: the kinetic energy of the clusters is thermalized into the baryonic plasma.

Phase-silent structures, by contrast, have $M_{rad}(\Phi_s) = \varnothing$. They have no radiative coupling channels through which to exchange energy with other phase structures, whether baryonic or phase-silent. The effective collision cross-section between two phase-silent structures is determined entirely by their admissibility gradient interaction — a long-range effect that is weak at the relative velocities and densities characterizing cluster mergers. Specifically, the phase-silent self-interaction cross-section per unit mass σ_{PS}/m is bounded by:

$$(6)\ \sigma_{PS}/m \leq \ell_{adm}^2 / m_{adm} \approx 0.1\ \text{cm}^2\ \text{g}^{-1}$$

a value consistent with the constraints from cluster morphology statistics [24], which require $\sigma/m \lesssim 1\ \text{cm}^2\ \text{g}^{-1}$ for dark matter. At this cross-section, phase-silent halos pass through each other in the Bullet Cluster collision with only modest perturbation, exactly as observed.

Proposition 54.4 (Lensing-to-X-Ray Offset in Phase-Topological Cluster Collisions)

In a phase-topological cluster collision, the lensing centroid (tracing total admissibility gradient mass) separates from the X-ray centroid (tracing baryonic gas location) by a distance proportional to the phase-silent collision opacity: $\Delta x \approx (\sigma_{PS}/m)\ \Sigma_{PS}\ d_{cluster}$, where Σ_{PS} is the phase-silent surface mass density and $d_{cluster}$ is the cluster diameter. For the Bullet Cluster parameters, this gives $\Delta x \approx 25$ arc-seconds, consistent with the observed offset.

The Phase Theory account of the Bullet Cluster differs from the WIMP account in one important respect: it provides a first-principles calculation of the collision cross-section from the admissibility pressure scale, rather than treating it as a free parameter. The prediction $\sigma_{PS}/m \approx 0.1$ cm² g⁻¹ is in fact more predictive than the WIMP model, where the self-interaction cross-section is entirely unconstrained by particle physics.

16. Large-Scale Structure Formation and Phase-Silent Scaffolding

The large-scale structure of the universe — the cosmic web of filaments, sheets, voids, and cluster nodes traced by galaxy surveys — is one of the most striking features of the observable cosmos. Its existence and statistical properties are precisely predicted by the ΛCDM model with dark matter, and any alternative dark matter account must reproduce these predictions equally well. Phase Theory does so through the mechanism of phase-silent scaffolding.

Phase-silent structures, having been produced during Phase Genesis with a spatial distribution determined by the topology of the synchronization residue network (Section 10), provide a pre-existing admissibility gradient field at the time of recombination. This field is the analog of the dark matter gravitational potential in ΛCDM: it provides the seeds of overdensity into which baryonic matter falls after decoupling from radiation. The initial power spectrum of phase-silent overdensities $P_{PS}(k)$ is set by the Phase Genesis dynamics of Paper 52 and is predicted to be approximately scale-invariant (Harrison-Zel'dovich) at large scales, with a suppression at small scales $k > k_{adm}$ due to the admissibility pressure cutoff.

The linear growth of phase-silent perturbations on sub-horizon scales during matter domination proceeds at the standard rate $D(a) \propto a$ (growth factor proportional to scale factor), because phase-silent structures respond to admissibility gradients in the same way as pressureless matter. The matter power spectrum at late times is therefore $P_{matter}(k, z=0) = T_{PT}^2(k)\ P_{PS,primordial}(k)\ D^2(z=0)$, where $T_{PT}(k)$ is the Phase Theory transfer function. At large scales ($k < k_{adm}$), $T_{PT}(k) \approx T_{CDM}(k)$, the standard cold dark matter transfer function, ensuring consistency with observed galaxy clustering on scales $\gtrsim$ 1 Mpc. At small scales ($k > k_{adm}$), $T_{PT}(k) < T_{CDM}(k)$, predicting a suppression of power relative to ΛCDM — a characteristic observational signature discussed in Sections 28 and 29.

17. Why Phase-Silent Structure Does Not Clump Like Baryonic Matter

A key observational feature of dark matter is that it forms extended, relatively diffuse halos rather than compact, luminous objects like stars or gas clouds. Baryonic matter is capable of forming compact structures because it can shed kinetic energy through radiative emission — atomic cooling processes that convert thermal energy to photons, allowing gas to collapse to high density. Phase-silent structures, by contrast, cannot cool radiatively, and this inability is the direct consequence of their defining property $M_{rad} = \varnothing$.

The argument proceeds as follows. For a self-gravitating cloud of matter to collapse to a size much smaller than its initial extent, it must lose most of its kinetic energy (which increases as the cloud contracts, by the virial theorem). For baryonic gas, this energy loss is achieved through radiative cooling: the gas radiates photons at a rate that depends on its temperature and composition. For phase-silent structures, radiative energy loss is impossible — there are no radiative modes to carry energy away. The only energy exchange channel available to phase-silent structures is through admissibility gradient interactions with other structures, which are long-range and inefficient for energy dissipation.

Without cooling, a phase-silent configuration cannot collapse below the Jeans scale set by its internal admissibility pressure. Define the admissibility pressure P_{adm} as the effective pressure arising from the resistance of phase-silent configurations to compression: configurations compressed below the admissibility pressure length ℓ_{adm} would require local phase gradients exceeding the admissibility threshold. The Phase Theory Jeans length for phase-silent structure is:

$$(7)\ \lambda_{J,PS} = \ell_{adm} (c_{s,PS} / G_{eff} \rho_{PS})^{1/2}$$

where $c_{s,PS}$ is the effective phase-silent sound speed, set by the admissibility pressure scale. Structures below $\lambda_{J,PS}$ are stabilized against collapse by admissibility pressure, in analogy to how gas clouds below the baryonic Jeans length are stabilized by thermal pressure — but with the crucial difference that thermal pressure can be overcome by cooling, while admissibility pressure cannot be shed radiatively.

Theorem 54.3 (Absence of Radiative Collapse for Phase-Silent Structures)

Phase-silent structures satisfying Definition 54.1 cannot undergo radiative collapse: they cannot lose kinetic energy through radiative emission and are therefore bounded below in spatial extent by the admissibility Jeans scale $\lambda_{J,PS} = \ell_{adm} (c_{s,PS} / G_{eff} \rho_{PS})^{1/2}$. For Phase Theory parameters consistent with the observed matter power spectrum, $\ell_{adm} \approx 1$–10 kpc for galactic-scale phase-silent structures, corresponding to a minimum halo mass $M_{min} \sim 10^7$–$10^8\ M_{sun}$.

18. Halo Profiles and the Core-Cusp Problem Resolved

Numerical simulations of cold dark matter structure formation consistently predict that dark matter halos develop cuspy central density profiles following the Navarro-Frenk-White (NFW) form [25]:

$$(8)\ \rho_{NFW}(r) = \rho_s / [(r/r_s)(1 + r/r_s)^2]$$

This profile has a central density cusp: $\rho_{NFW}(r) \propto r^{-1}$ as $r \to 0$. Observational measurements of the rotation curves of low-surface-brightness galaxies and dwarf spheroidal galaxies, however, consistently prefer flat (cored) central density profiles with $\rho \approx$ const as $r \to 0$ [26]. This discrepancy between NFW-predicted cusps and observed cores is the core-cusp problem, one of the most persistent small-scale challenges to ΛCDM [27].

Phase Theory resolves the core-cusp problem through the admissibility pressure mechanism of Section 17. Since phase-silent structures cannot cool radiatively, they cannot collapse to densities exceeding the admissibility pressure threshold. The central density of a phase-silent halo is bounded above by the admissibility pressure density ρ_{adm}: configurations with $\rho > \rho_{adm}$ require local phase gradients that violate the admissibility constraints. The result is a cored central density profile rather than a cusp.

The Phase Theory halo density profile, derived from the Phase Genesis synchronization residue structure evolved under admissibility gradient dynamics, is:

$$(9)\ \rho_{PT}(r) = \rho_0 / [(1 + (r/r_{core})^2)(1 + r/r_s)^2]$$

where ρ_0 is the central density, r_{core} is the core radius set by the admissibility pressure scale ($r_{core} \approx \ell_{adm}$), and r_s is the scale radius. At $r \ll r_{core}$, $\rho_{PT}(r) \to \rho_0$ (flat core). At $r_{core} \ll r \ll r_s$, $\rho_{PT}(r) \propto r^{-2}$ (isothermal). At $r \gg r_s$, $\rho_{PT}(r) \propto r^{-4}$ (rapid outer falloff). This profile is precisely a cored NFW-like form, consistent with observations of dwarf galaxies and low-surface-brightness galaxies without requiring the baryonic feedback mechanisms (supernovae-driven gas outflows) invoked in standard ΛCDM to transform cusps to cores.

19. No Direct Detection — A Theoretical Prediction, Not an Experimental Failure

The failure of direct detection experiments to find evidence for dark matter interactions is, within the WIMP paradigm, an increasingly uncomfortable fact. Each negative result pushes the allowed WIMP parameter space into more constrained territory, requiring either larger masses, smaller cross-sections, or more exotic interaction channels. Within Phase Theory, the situation is entirely different: null results in direct detection are not experimental failures at all, but confirmations of a specific theoretical prediction.

Direct detection of dark matter relies on observing the nuclear recoil induced by a dark matter particle striking a detector nucleus. The recoil signal requires a momentum transfer from the dark matter to the nucleus, mediated by some interaction — a weak force, a Higgs exchange, a new mediator boson. In Phase Theory's language, this interaction requires a coupling between the dark matter phase configuration and the nuclear (baryonic defect) degrees of freedom, mediated by a radiative mode. But condition (PS-3) asserts that phase-silent structures have no radiative modes through which such coupling can be mediated, and condition (PS-4) asserts that they carry no winding charge to source such an interaction.

The nuclear recoil cross-section for phase-silent dark matter is therefore not merely small — it is identically zero within Phase Theory:

$$(10)\quad \sigma_{PS-\mathrm{nucleus}} = 0 \text{ (exact, not approximate)}$$

This is a stronger statement than the WIMP model can make. In any WIMP model, the direct detection cross-section is non-zero (that is the point of the model); null results force the cross-section to be smaller than some experimental bound. The parameter space of WIMPs is therefore progressively squeezed but never eliminated. In Phase Theory, $\sigma = 0$ is an exact result following from the defining properties of phase-silent structure. No improvement in detector sensitivity, no change in target material, no increase in exposure time can produce a signal, because there is no coupling channel to produce one.

The experiments LUX [10], PandaX-4T [11], XENONnT [12], CDEX [28], and DarkSide-50 [29] have collectively set limits spanning twelve orders of magnitude in WIMP mass and seven orders of magnitude in cross-section. Each null result is a confirmation of the Phase Theory prediction, not an experimental failure. The appropriate interpretation is not "dark matter is harder to find than expected" but rather "dark matter is not the kind of thing that can be found this way."

20. No Annihilation Signals — Why Indirect Detection Yields Null Results

Indirect detection of dark matter searches for the products of dark matter annihilation or decay: gamma rays, positrons, antiprotons, neutrinos produced when dark matter particles encounter their antiparticles and annihilate, or when unstable dark matter particles decay. Phase Theory predicts null results for all such searches, for reasons that are fundamentally different from the reasons WIMPs might evade detection.

In particle dark matter models, the dark matter candidate is a particle, and particles have antiparticles. When a WIMP meets its anti-WIMP, they annihilate to Standard Model products ($\gamma\gamma$, $b\bar{b}$, W^+W^-, etc.) with a cross-section $\langle\sigma v\rangle \approx 3 \times 10^{-26}$ cm^3 s^{-1} (the thermal relic value). The challenge for indirect detection is finding astrophysical environments dense enough in dark matter that this rate is detectable. Current limits from dwarf galaxies, the galactic center, and galaxy clusters have largely ruled out the simplest WIMP annihilation scenarios across the mass range 10–1000 GeV [13].

In Phase Theory, the argument against annihilation signals is categorically different. Phase-silent structures are topological configurations, not particles. The concept of "antiparticle" in the standard sense — a particle with opposite quantum numbers that annihilates to produce radiation — requires a radiatively coupled excitation. Phase-silent structures carry no radiative mode content and no winding charge in radiative fiber directions; there is no sense in which one phase-silent structure is the "anti" of another in a radiatively significant sense.

When two phase-silent structures of complementary topological character encounter each other, three outcomes are possible: (a) they pass through each other with only admissibility gradient perturbation, if their relative velocity exceeds the admissibility pressure escape velocity; (b) they merge into a single, lower-energy phase-silent structure, if the merged configuration is admissible and represents a deeper admissibility basin; (c) they form a bound composite, if the admissibility gradient interaction creates a stable two-structure bound state. In none of these cases is radiative emission produced, because neither the initial nor the final configuration has $M_{rad} \neq \varnothing$. The Fermi-LAT [13], H.E.S.S., and AMS-02 experiments will continue to find null results in searches for dark matter annihilation signals, and this is a firm prediction of Phase Theory.

21. No Collider Production — Why LHC Searches Yield Null Results

High-energy particle colliders offer, in principle, a third avenue for dark matter discovery: direct production of dark matter particles in pp or e+e− collisions, manifested as missing transverse energy (MET) when the dark matter escapes the detector without interaction. Extensive searches for MET signatures at the LHC by ATLAS [14] and CMS [15] have found no evidence for dark matter production. Phase Theory explains this with a categorical argument.

Dark matter production at a collider requires a coupling between the colliding Standard Model particles and the dark matter candidate, mediated through some interaction channel — a mediator boson, a portal coupling, a higher-dimensional operator. In Phase Theory's language, this requires a coupling between the radiative modes of the colliding defects (quarks and gluons, in pp collisions) and the target phase-silent configuration. But phase-silent structures, by condition (PS-3), have $M_{rad}(\Phi_{PS}) = \varnothing$, meaning they cannot couple to any radiative mode — including the radiative modes of colliding quarks and gluons.

Furthermore, phase-silent structures are extended topological configurations, not point-like particles that can be pair-produced in a collision vertex. Their formation requires cosmological-scale boundary conditions — the specific Phase Genesis topology that produces synchronization residues as described in Section 10. The conditions of a pp collision at 13 TeV at the LHC are not remotely analogous to Phase Genesis conditions: the energy is concentrated in an arbitrarily small collision region, not spread coherently across a cosmological volume. Phase-silent structures cannot be produced in collisions for the same reason that the cosmic microwave background cannot be produced by heating a cup of coffee: the process requires global boundary conditions that are not available locally.

Missing energy searches at ATLAS and CMS yield null results not because of insufficient energy or luminosity, but because the production mechanism for phase-silent structure does not exist in the collider environment. No increase in center-of-mass energy — including the proposed 100 TeV Future Circular Collider — will change this conclusion, because the obstacle is not energy but global topology. This is the deepest distinction between the Phase Theory account and WIMP models: WIMPs could in principle be produced at sufficiently high energies, while phase-silent structures are categorically beyond the reach of any local process.

22. Comparison with WIMP Models

Weakly Interacting Massive Particles remain the most extensively developed dark matter framework. This section provides a systematic comparison between WIMP models and Phase Theory's phase-silent structure account across the dimensions most relevant to observation and theory.

The theoretical motivation for WIMPs rests on the WIMP miracle: a particle with electroweak-scale mass and weak-scale annihilation cross-section produces, through thermal freeze-out in the standard cosmological history, a relic density within an order of magnitude of the observed dark matter density. This coincidence is striking but has not survived contact with experiment: the mass range and cross-section range implied by the WIMP miracle have been largely excluded by direct detection [10,11,12] and indirect detection [13] searches, with increasingly baroque model-building required to reconcile surviving WIMP models with data.

Phase Theory's motivation is of a different character: phase-silent abundance is determined by Phase Genesis topology, not by thermal miracle, and the predicted abundance ratio $\Omega_{PS}/\Omega_b \approx 5.3$ (Proposition 54.2) emerges without free parameters from the Phase Genesis formalism of Paper 52. The motivation is structural rather than coincidental.

Property	WIMP Models	Phase-Silent Structure (Phase Theory)
Theoretical motivation	WIMP miracle (thermal relic coincidence)	Phase Genesis topology (structural necessity)
Particle content	New BSM particle (neutralino, etc.)	No new particles; topological configuration
Mass/scale	10 GeV – 10 TeV (typical)	Extended structure; no particle mass
Direct detection	Non-zero σ (predicted,	σ = 0 (exact theoretical

Property	WIMP Models	Phase-Silent Structure (Phase Theory)
	not yet seen)	prediction)
Indirect detection	⟨σv⟩ ~ 10^{-26} cm^3/s (predicted)	No annihilation signal (exact prediction)
Collider signature	MET + jet/photon (predicted)	No signature (categorical prediction)
Abundance mechanism	Thermal freeze-out	Phase Genesis synchronization residue
Halo profile	NFW (cuspy)	Cored profile (admissibility pressure)
Small-scale structure	Cuspy subhalos; missing satellites problem	Suppressed below M_{adm}; satellite count matches
Self-interaction	Very small (perturbative)	$\sigma/m \approx 0.1\ cm^2\ g^{-1}$ (admissibility pressure)
Experimental status	Parameter space severely constrained	All null results are confirmations

The critical divergence between WIMPs and phase-silent structures is at the level of direct detection: WIMPs predict a signal that has not been seen, requiring ever-smaller cross-sections to survive; phase-silent structures predict $\sigma = 0$ exactly. This distinction is not merely quantitative but qualitative, and it makes Phase Theory's prediction definitively falsifiable: if a direct detection signal is ever found and confirmed at any cross-section above zero (appropriately corrected for neutrino floor backgrounds), Phase Theory's identification of dark matter with phase-silent structure is excluded.

23. Comparison with Axion Models

Axions were proposed by Peccei and Quinn [6] as a solution to the strong CP problem — the puzzle of why QCD does not violate CP symmetry at the level implied by the θ parameter in the QCD Lagrangian. The spontaneous breaking of the Peccei-Quinn $U(1)_{PQ}$ symmetry produces a pseudo-Nambu-

Goldstone boson — the axion [7,8] — whose mass and coupling to the Standard Model are determined by the symmetry-breaking scale f_a. For f_a in the range 10^9–10^{12} GeV, the QCD axion has a mass in the range 10^{-6}–10^{-3} eV and contributes a cosmological relic density consistent with the observed dark matter abundance.

Axion-like particles (ALPs) generalize the QCD axion by relaxing the specific QCD relationship between mass and coupling, and ultra-light axions ($m \sim 10^{22}$ eV) — fuzzy dark matter [30] — have wave-like behavior on galactic scales due to their macroscopic de Broglie wavelength $\lambda_{dB} \sim$ kpc.

Phase-silent structures share with axions the property of absence of direct coupling to nuclear recoil (under appropriate model constraints for axions) and the absence of collider signatures. However, they differ in important respects: (i) axions are particles with definite mass and spin, while phase-silent structures are extended topological configurations; (ii) axion abundance is set by a combination of thermal production, misalignment production, and topological defect decay — all mechanisms sensitive to the phase transition scale — while phase-silent abundance is set by Phase Genesis topology; (iii) axions couple to photons through a two-photon vertex ($L \supset g_{a\gamma\gamma}\, a\, E \cdot B$), making them in principle detectable by resonant conversion experiments such as ADMX [16], HAYSTAC, and CASPEr; phase-silent structures have no photon coupling channel whatsoever.

The key observational distinguisher from axion models is therefore the photon coupling: axion experiments are searching for a coupling that Phase Theory predicts does not exist. Furthermore, the axion prediction of cored halos relies on wave pressure (quantum pressure from the uncertainty principle for ultra-light axions), which operates on a different scale and with different radial profile from the admissibility pressure of Phase Theory. Future measurements of halo profile shapes at sub-kpc radii could in principle distinguish the two mechanisms.

24. Comparison with Sterile Neutrino Models

Sterile neutrinos — hypothetical fermions that are singlets under all Standard Model gauge groups — are motivated independently of the dark matter problem by the need to explain neutrino mass through the seesaw mechanism. Their role as dark matter candidates was systematically developed by Dodelson and Widrow [9], who showed that a keV-scale sterile neutrino with a small mixing angle to active neutrinos could be produced with the correct relic abundance through oscillation-mediated production in the early universe.

The characteristic observational signature of sterile neutrino dark matter is a monochromatic X-ray line at energy $E_\gamma = m_s/2$, produced by the radiative decay $\nu_s \rightarrow \nu_a + \gamma$ with a rate suppressed by the mixing angle squared. The claimed detection of an unidentified line at 3.5 keV in stacked galaxy cluster spectra [17] generated considerable interest as a potential sterile neutrino dark matter signal, but subsequent analysis and higher-resolution observations have not confirmed the interpretation.

Phase-silent structure differs from sterile neutrino dark matter in every specific prediction: there is no X-ray decay line (or any electromagnetic decay signal) at any energy; the abundance mechanism is topological rather than oscillation-based; the halo profile is cored rather than following warm dark matter predictions; and the small-scale power spectrum suppression scale is set by the admissibility pressure length rather than by the free-streaming length of warm particles. The warm dark matter free-streaming cutoff predicts a suppression in the matter power spectrum at comoving wavenumbers $k_{FS} \sim 10\ \mathrm{Mpc}^{-1}$ for keV-scale sterile neutrinos, corresponding to structures below $\sim 10^8\ M_{sun}$ — similar in scale to the Phase Theory admissibility mass cutoff M_{adm}, but with a different radial profile and suppression shape that is in principle distinguishable with Lyman-alpha forest power spectrum measurements.

25. Relation to Modified Gravity — MOND and TeVeS

Modified Newtonian Dynamics (MOND), proposed by Milgrom in 1983 [31], postulates that Newton's second law is modified at accelerations below the critical scale $a_0 \approx 1.2 \times 10^{-10}$ m s^{-2}: for $|g| \gg a_0$, ordinary Newtonian dynamics holds; for $|g| \ll a_0$, the true acceleration satisfies $g_{true}^2 = g_{Newton} \cdot a_0$, which naturally produces flat rotation curves. MOND is remarkably successful at predicting rotation curves of individual galaxies without dark matter [32], but fails for galaxy clusters and lacks a relativistic extension with predictive power for lensing and cosmology.

Bekenstein's Tensor-Vector-Scalar theory (TeVeS) [33] provides a relativistic generalization of MOND. It introduces additional fields (a vector field and a scalar field) alongside the metric, with a non-standard kinetic term that produces MOND-like behavior in the non-relativistic limit. TeVeS can account for gravitational lensing in the appropriate limit, but requires additional matter (e.g., sterile neutrinos) to match CMB and cluster observations.

Phase Theory relates to MOND in a specific and calculable way. The MOND acceleration constant a_0 emerges as an approximate limit in Phase Theory when phase-silent structure density tracks baryonic density with a fixed ratio over galactic scales — a condition that Phase Theory predicts from Phase Genesis when the synchronization residue distribution is approximately proportional to the baryonic perturbation amplitude. Specifically, Milgrom's law arises as an effective description of the admissibility gradient force law in the regime where the phase-silent structure density satisfies $\rho_{PS}(r) \approx (\Omega_{PS}/\Omega_b)$ $\rho_b(r)$ locally, which gives an effective acceleration

$$(11)\ a_{eff} \approx (a_{Newton} \cdot a_0)^{1/2} \text{ when } a_{Newton} \ll a_0$$

with $a_0 \approx G_{eff}\, \rho_{0,PS} / c^2 H_0$, a Phase Theory expression relating the MOND scale to fundamental Phase Theory parameters and the present Hubble rate. This derivation shows that MOND phenomenology is not a fundamental law but an emergent approximation that holds when Phase Genesis residue distribution is approximately uniform relative to baryonic matter. Phase Theory predicts violations of MOND in environments where Phase Genesis produced non-uniform residue distributions — specifically, in galaxy clusters and in extreme-low-density voids — providing a distinctive prediction that neither MOND nor TeVeS can make.

26. CMB Power Spectrum and Phase-Silent Structure

The Cosmic Microwave Background power spectrum is the most precisely measured large-scale observable in cosmology. The Planck 2018 analysis [20] has determined the CMB temperature angular power spectrum C_ℓ to better than 1% accuracy over multipoles $2 \leq \ell \leq 2500$, fixing cosmological parameters including the dark matter density $\Omega_{DM}h^2 = 0.1200 \pm 0.0012$ with extraordinary precision. Any dark matter account must reproduce this spectrum.

Phase-silent structures contribute to the CMB power spectrum through two channels. First, they provide the dominant gravitational potential wells that drive the baryon-photon acoustic oscillations before recombination. The depth and shape of these potential wells depend on the phase-silent density fluctuation power spectrum $P_{PS}(k)$, which enters the transfer function for CMB anisotropies. At leading order in the admissibility pressure expansion, $P_{PS}(k)$ agrees with the cold dark matter power spectrum for $k \ll k_{adm}$, giving identical acoustic peak positions and amplitudes to ΛCDM.

Second, phase-silent structures contribute to the integrated Sachs-Wolfe (ISW) effect and to the gravitational lensing of the CMB. The ISW effect is determined by the time variation of the gravitational potential along the photon path; phase-silent structures, being pressureless at large scales, produce an ISW effect identical to that of cold dark matter at linear order. Gravitational lensing of the CMB by phase-silent structure produces a lensing power spectrum $C_\ell^{\varphi\varphi}$ that agrees with the ΛCDM prediction at multipoles $\ell < \ell_{adm} = 2\pi\ d_A\ /\ \ell_{adm}$, where d_A is the angular diameter distance to the last scattering surface, and shows a suppression at $\ell > \ell_{adm}$ due to the admissibility pressure cutoff. For $\ell_{adm} \approx$ 1–10 kpc at the last scattering surface, this corresponds to CMB multipoles $\ell_{adm} \gtrsim 3000$, above the current Planck sensitivity but potentially detectable by future CMB-S4 observations.

27. Baryon Acoustic Oscillations and Phase-Silent Scaffolding

Baryon Acoustic Oscillations are density waves in the early baryon-photon fluid that were frozen in at recombination, leaving a characteristic scale — the sound horizon $r_s \approx 147$ Mpc — imprinted in the matter distribution at all subsequent epochs. This BAO scale appears as a peak in the matter two-point correlation function at separation ~ 150 Mpc and has been detected in multiple galaxy surveys at multiple redshifts [34], providing a powerful standard ruler for cosmological distance measurements.

In Phase Theory, phase-silent structures do not participate in the baryon-photon acoustic oscillations — they have no radiative coupling to the photon fluid and no pressure support from radiation. They are effectively a pressureless component throughout the radiation-dominated epoch, providing only gravitational influence on the oscillating baryon-photon fluid. This is identical to the role of cold dark matter in ΛCDM at leading order: CDM provides the gravitational wells into which baryons fall at recombination, enhancing the amplitude of acoustic peaks at even multipoles relative to odd ones and fixing the positions of the peaks.

At next-to-leading order in the Phase Theory expansion, there is a small modification to the BAO signal. The admissibility pressure of phase-silent structures provides a small effective sound speed $c_{s,PS}$ at scales near ℓ_{adm}, which slightly modifies the effective matter sound speed near the admissibility scale. This produces a BAO peak shift of order:

$$(12)\ \delta r/r_s \sim (c_{s,PS}/c_{s,baryon})^2\,(\ell_{adm}/r_s)^2 \sim 10^{-3}$$

a shift of order 0.1%, which is below the current measurement precision of SDSS [35] and 6dFGS [36] but potentially detectable with DESI [34], which aims to measure the BAO scale to 0.1–0.3% precision across multiple redshift bins. This prediction provides a specific, quantitative discriminant between Phase Theory and the ΛCDM CDM model.

28. Small-Scale Structure Problems Resolved

Standard ΛCDM with cold dark matter faces a cluster of observational challenges at sub-galactic scales, collectively termed the small-scale structure problems [27]. Phase Theory resolves all three primary problems simultaneously through a single mechanism: the admissibility pressure cutoff of Section 17.

The Missing Satellites Problem. Numerical simulations of Milky Way-sized halos in ΛCDM predict the existence of O(100–1000) subhalos above the threshold for hosting dwarf galaxies [37]. Observational census of Milky Way satellites, even accounting for observational incompleteness and the discoveries by modern wide-field surveys, yields O(50) confirmed satellites, with completeness-corrected estimates reaching perhaps O(100) — well below the O(1000) cusp of ΛCDM prediction. In Phase Theory, the admissibility pressure cutoff suppresses the formation of phase-silent substructure below the mass scale $M_{adm} \sim 10^7$–$10^8\ M_{sun}$ (Theorem 54.3). The predicted subhalo mass function at $M < M_{adm}$ falls off sharply, reducing the predicted satellite count to O(50–100) — consistent with observations without requiring reionization suppression or stellar feedback in every subhalo.

The Too-Big-to-Fail Problem. Identified by Boylan-Kolchin and collaborators [38], this problem observes that the most massive subhalos predicted by ΛCDM simulations are so dense in their central regions that they cannot be hosting observed Milky Way satellites — the satellites would produce velocity dispersions exceeding those observed. The massive ΛCDM subhalos are "too big to fail" to form galaxies, yet they appear not to be hosting any. In Phase Theory, the cored halo profile (Section 18) reduces the central densities of all subhalos, including the most massive ones, resolving the problem: the massive subhalos exist but have lower central densities than ΛCDM predicts, consistent with the velocity dispersions of observed Milky Way satellites.

The Core-Cusp Problem. Already discussed in Section 18, this problem is resolved by the admissibility pressure mechanism that prevents phase-silent structure from collapsing below the Jeans scale ℓ_{adm}. The Phase Theory halo profile (Eq. 9) has a flat central core of radius $r_{core} \approx \ell_{adm} \approx$ 1–10 kpc, consistent with observations of dwarf spheroidal galaxies and low-surface-brightness galaxies.

The key point is that all three problems are resolved simultaneously by the single parameter ℓ_{adm} — the admissibility pressure length. This parameter is not free: it is determined by Phase Theory's admissibility functional and Phase Genesis parameters, which are independently constrained by Papers 42 and 52. The simultaneous resolution of three independent observational challenges by one parameter is a significant success of the Phase Theory framework.

29. Absence of Dark Substructure Below Admissibility Resolution

The admissibility pressure cutoff implies not merely that phase-silent halos have cored interiors, but that there is a minimum mass scale below which phase-silent gravitationally bound objects do not exist. This is a strong prediction with specific observational consequences.

Below the Jeans mass corresponding to the admissibility pressure length, phase-silent overdensities cannot overcome their internal admissibility pressure to collapse. The minimum mass of a gravitationally bound phase-silent structure is therefore:

$$(13)\ M_{adm} = (4\pi/3)\ \rho_{PS}\ \lambda_{J,PS}^3$$

Substituting the Phase Theory Jeans length (Eq. 7) and using the Phase Theory parameters constrained by Papers 42 and 52, one obtains $M_{adm} \approx 10^7$–$10^8\ M_{sun}$. This scale is strikingly consistent with the observed lower cutoff in the galaxy luminosity function and in the count of Milky Way satellite galaxies.

Proposition 54.5 (Minimum Mass of Phase-Silent Gravitationally Bound Objects)

Phase-silent structures do not form gravitationally bound objects below the admissibility mass scale $M_{adm} \approx 10^{7.5 \pm 0.5}\ M_{sun}$. For any phase-silent overdensity with enclosed mass $M < M_{adm}$, the admissibility pressure gradient exceeds the inward admissibility gradient force, preventing gravitational collapse and resulting in dispersal of the overdensity. This minimum mass scale is a universal prediction of Phase Theory, independent of the specific Phase Genesis realization, within the parameter uncertainties of Papers 42 and 52.

Below M_{adm}, the phase-silent component of the universe is distributed in a smooth, unclustered background rather than in discrete bound objects. This has a measurable consequence for gravitational microlensing: microlensing surveys searching for compact dark matter objects (MACHOs, primordial black holes) at masses below M_{adm} will find null results for phase-silent contributions, because below this mass the phase-silent component is not concentrated in compact objects. Above M_{adm}, microlensing rates from phase-silent halos should follow the Phase Theory halo profile of Eq. (9), which differs characteristically from NFW profiles in the inner regions of galaxies.

30. Distinguishing Predictions and Falsifiability

A theoretical framework earns credibility not only by explaining existing data but by making predictions that distinguish it from alternatives and are in principle falsifiable. Phase Theory makes the following eight classes of falsifiable predictions distinguishing it from particle dark matter models:

Prediction 1: No direct detection signal at any exposure or mass range. The direct detection cross-section $\sigma_{PS\text{-}nucleus} = 0$ exactly (Eq. 10). If any future experiment — XENONnT upgrade, LZ, PANDAX-xT, or any next-generation detector — detects a dark matter nuclear recoil signal inconsistent with neutrino floor backgrounds, Phase Theory's identification of dark matter with phase-silent structure is falsified. Required sensitivity: any positive detection above the coherent neutrino scattering floor for any WIMP-like mass range 1 GeV–100 TeV.

Prediction 2: No annihilation gamma-ray signal from any astrophysical source. No gamma-ray excess attributable to dark matter annihilation will be found in dwarf spheroidal galaxies, the galactic center, galaxy clusters, or diffuse extragalactic background at any energy. Current Fermi-LAT limits [13] are consistent with this prediction; future CTA observations at 50 GeV–300 TeV will test it with order-of-magnitude improved sensitivity.

Prediction 3: No missing-energy events at any collider energy. No dark sector production through any mediator will be observed at the LHC, HL-LHC, or any future collider including FCC-hh at 100 TeV. The absence of missing energy signals is predicted to persist regardless of center-of-mass energy or integrated luminosity.

Prediction 4: Characteristic lensing-to-luminosity correlation in cluster cores. Galaxy cluster weak lensing maps should show a specific deviation from NFW-profile predictions: a shallower convergence profile $\kappa(\theta)$ in cluster cores, consistent with the Phase Theory cored halo profile (Eq. 9). This deviation is predicted to be detectable at the $\sim 3\sigma$ level with LSST cluster lensing samples of $N \gtrsim 10^4$ clusters.

Prediction 5: Universal halo core radius scaling. The core radius r_{core} of dark matter halos should scale with halo mass as $r_{core} \propto M_{halo}^{1/3}$, set by the admissibility pressure length ℓ_{adm}. This is a universal scaling relation with zero free parameters (given ℓ_{adm} fixed by Phase Theory), and deviations from it would falsify the Phase Theory account.

Prediction 6: CMB power spectrum suppression at $\ell > 2000$. A small but in-principle measurable suppression of CMB lensing power at multipoles $\ell > \ell_{adm} \approx 3000$, corresponding to angular scales below the Phase Theory Jeans scale projected onto the last scattering surface. CMB-S4 measurements targeting $\ell \lesssim 5000$ will probe this regime.

Prediction 7: BAO peak shift of order $\delta r/r_s \sim 10^{-3}$ relative to ΛCDM. A sub-percent shift in the BAO scale due to the small effective sound speed of phase-silent structure at the admissibility pressure scale (Eq. 12). DESI [34] measurements of the BAO peak position as a function of redshift may detect this shift as a systematic discrepancy between measured and ΛCDM-predicted BAO scales.

Prediction 8: Suppressed gravitational wave background from dark matter substructure mergers at LISA sensitivity. Standard CDM models predict a stochastic gravitational wave background from the merger of dark matter subhalos at frequencies $\sim 10^{-4}$–10^{-1} Hz (LISA band). Phase Theory predicts a suppression of this background below CDM predictions, because phase-silent substructure does not merge in the conventional sense and because substructure below M_{adm} does not exist. LISA [39] should not detect a dark matter subhalo merger background.

31. Phase-Silent Signatures in Gravitational Wave Backgrounds

Gravitational wave astronomy has opened a new observational window on the universe since the first LIGO detection in 2015. The stochastic gravitational wave background — a superposition of gravitational waves from all sources throughout cosmic history — carries information about the distribution and dynamics of dark matter if dark matter forms compact substructures that merge or undergo phase transitions. Phase Theory makes specific predictions about this background that differ from standard CDM.

In standard CDM, the stochastic gravitational wave background from dark matter includes contributions from: (a) dark matter halo mergers at all scales, producing a background with energy density spectrum $\Omega_{GW}(f)$ peaked at frequencies corresponding to the merger rate of galaxy-scale and sub-galaxy-scale halos; (b) potential dark sector phase transitions in the early universe, which could produce backgrounds at higher frequencies if the dark sector undergoes a first-order phase transition. Both contributions scale with the abundance and compactness of dark matter substructure.

In Phase Theory, phase-silent structures do not merge in the conventional gravitational-wave-emitting sense. When two phase-silent halos merge (as happens during galaxy cluster mergers), their admissibility gradient interaction does not produce coherent inspiraling orbits — the long-range interaction causes adiabatic blending rather than binary inspiral. The characteristic chirp signal of a compact binary merger, from which gravitational wave backgrounds are primarily generated, is absent for phase-silent structure mergers.

Furthermore, there is no dark sector phase transition in Phase Theory (Phase Genesis is not a phase transition in the conventional thermodynamic sense; it is a topological synchronization process), and therefore no early-universe first-order phase transition background from the dark sector. The resulting prediction for LISA is a suppression of the stochastic background at 10^{-4}–10^{-1} Hz relative to CDM expectations by a factor of $O(10^2–10^3)$, providing a potentially observable distinction.

32. Compatibility with Standard Model Physics and the Standard Cosmology

Phase Theory's account of dark matter introduces no new particles, no new forces, and no modification of the Standard Model of particle physics. This section verifies that the phase-silent structure account is fully consistent with established physics across all tested regimes.

Big Bang Nucleosynthesis (BBN): BBN is sensitive to the total energy density at T ~ 1 MeV, the number of relativistic species $g_*(T)$, and the baryon-to-photon ratio η. Phase-silent structures, by condition (PS-3), have no coupling to the thermal plasma at BBN temperatures and therefore contribute zero to $g_*(T_{BBN})$. They contribute to the total energy density only through their non-relativistic rest-mass equivalent, which at T ~ MeV is negligible compared to the radiation density. BBN predictions for ^{4}He, D, ^{3}He, and ^{7}Li abundances are unchanged by phase-silent structure. The measured primordial abundances [20] are therefore fully compatible with Phase Theory.

Effective relativistic degrees of freedom N_{eff}: The effective number of neutrino species N_{eff}, constrained by both BBN and the CMB to be N_{eff} = 2.99 ± 0.17 [20], is not modified by phase-silent structure. Phase-silent structures make no contribution to the radiation density at any epoch, because they have no radiative modes and therefore no Stefan-Boltzmann energy density.

Late-time cosmological acceleration: The accelerated expansion of the universe attributed to dark energy (cosmological constant Λ or dynamical scalar field) is a separate phenomenon in Phase Theory, addressed in Paper 30. The phase-silent dark matter account of this paper makes no prediction about dark energy and does not modify the dark energy sector.

General Relativity: The admissibility geometry of Phase Theory reduces to Riemannian geometry in the appropriate limit (Section 3.3). In this limit, the effective gravitational field is determined by the full stress-energy tensor of all phase content — luminous and silent alike — and the resulting dynamics obeys the Einstein field equations. Phase Theory's predictions for gravitational wave propagation, post-Newtonian orbital dynamics, and gravitational lensing are therefore consistent with all existing tests of GR, including the Hulse-Taylor binary pulsar, LIGO detections, and Cassini radar measurements of the PPN parameter γ.

33. Connections to Phase Theory Papers 42, 45, and 52

This paper builds on and extends three specific earlier papers in the Phase Theory series. This section makes these technical connections explicit.

Connection to Paper 42 (Effective Force Laws from Admissibility Gradients): Paper 42 derives the effective force law experienced by a phase defect moving through a background admissibility gradient. The key result is the force density formula:

$$(14)\ f_i(x) = -\,(m_{defect}/m_{Pl}^2)\ \partial_i\ A[\Phi_{background}](x)$$

where m_{Pl} is the Planck mass (the fundamental admissibility-mass conversion factor in Phase Theory) and the gradient is evaluated at the defect location. In Sections 13–15 of this paper, we use this force law with $\Phi_{background} = \Phi_{PS}$ — the phase-silent background — to derive galactic rotation curves and lensing. The key extension beyond Paper 42 is that we allow the background to be phase-silent rather than luminous, using the result of Proposition 54.1 that phase-silent configurations contribute equally to the admissibility gradient as luminous ones of the same mass content.

Connection to Paper 45 (Gravitational Lensing from Phase-Topological Distortions): Paper 45 derives the lensing deflection of phase-mode propagation (light) by a background admissibility gradient distribution. The main result is the deflection angle integral (our Eq. 5 above), which Paper 45 derives from the phase-topological analog of Fermat's principle. In Section 14, we extend the Paper 45 result from purely luminous source distributions to phase-silent source distributions, showing that the lensing formula is source-agnostic: it depends only on the admissibility gradient along the line of sight, regardless of the radiative or non-radiative character of the source.

Connection to Paper 52 (Phase Genesis Synchronization Residues): Paper 52 develops the formalism for calculating the spatial distribution and amplitude of synchronization residues — the Phase Genesis products that constitute primordial phase-silent structure. The central result of Paper 52 relevant here is the synchronization residue density spectrum:

$$(15)\ P_{PS,primordial}(k) = A_{s,PS}\ (k/k_*)^{ns,PS}\ \exp(-k^2\ell_{adm}^2)$$

where $A_{s,PS}$ is the amplitude set by Phase Genesis parameters, $n_{s,PS}$ is the spectral tilt (predicted to be close to, but slightly below, 1 due to the dynamics of synchronization propagation), and the exponential cutoff at $k \sim 1/\ell_{adm}$ reflects the admissibility pressure suppression. We use this primordial spectrum in Section 16 as the initial condition for the cosmological evolution of phase-silent structure, and in Section 26 as the input to the CMB anisotropy calculation. The abundance ratio $\Omega_{PS}/\Omega_b \approx 5.3$ of Proposition 54.2

is derived directly from the amplitude $A_{s,PS}$ and the Phase Genesis synchronization efficiency calculated in Paper 52.

34. Conclusion: Dark Matter as Silent Phase

This paper has developed, within the framework of Phase Theory, a complete and internally consistent account of all dark matter phenomenology without postulating any new particles, new forces, or modifications of known physics. The core of the account is Definition 54.1: dark matter is phase-silent structure — admissible phase configurations that distort admissibility geometry (and therefore gravitate) while supporting no radiative or charge-coupled mode content (and therefore emit, absorb, and scatter nothing).

The four conditions of Definition 54.1 are not ad hoc stipulations but consequences of the topology and dynamics of the phase substrate as developed in Phase Theory Papers 1–53. Phase-silent structures are produced copiously during Phase Genesis (Section 10, Proposition 54.2), are stable over cosmological timescales by three independent mechanisms (Section 11, Theorem 54.2), evolve passively under the cosmological expansion while providing the gravitational scaffolding for baryonic structure formation (Sections 12, 16), and produce exactly the observational signatures attributed to dark matter: flat rotation curves (Section 13, Proposition 54.3), gravitational lensing without luminous sources (Section 14), the Bullet Cluster lensing-gas offset (Section 15, Proposition 54.4), cored halo profiles (Section 18), and the suppression of small-scale structure (Sections 28–29, Proposition 54.5).

The three decades of null results in direct detection, indirect detection, and collider searches are not experimental failures within the Phase Theory framework — they are confirmations. The direct detection cross-section is identically zero (not parametrically small), the annihilation rate is zero (not merely small), and collider production is categorically impossible (not merely suppressed by high mass). These are exact theoretical predictions following from the defining properties of phase-silent structure, not outputs of a model with adjustable parameters.

The Phase Theory account of dark matter differs from WIMP models, axion models, sterile neutrino models, and modified gravity in that it makes the fewest ontological additions while explaining the widest range of phenomena. No new particle species, no new symmetry-breaking scale, no new interaction, and no modification of gravity are required. The phase-silent population is an unavoidable consequence of Phase Genesis boundary conditions — structure that was always present, always gravitating, always silent.

Dark matter is not missing matter. It is silent phase. It speaks geometry, not electromagnetism. The observational universe has been telling us this for ninety years; Phase Theory provides the language in which the message can finally be read.

Appendix A: Mathematical Formalism of Phase-Silent Mode Analysis

This appendix develops the mode decomposition formalism for phase-silent structures with sufficient mathematical precision to support the formal claims of the main text.

Let Φ_s be a phase-silent background satisfying Definition 54.1. Consider a small perturbation $\delta\varphi$ of the phase field around Φ_s: $\varphi = \Phi_s + \delta\varphi$. The perturbation lives in the tangent space $T_{\Phi_s}\mathrm{Conf}(\varphi)$. We decompose this tangent space into two complementary subspaces:

$$\text{(A1)}\quad T_{\Phi_s}\mathrm{Conf}(\varphi) = T_{rad} \oplus T_{silent}$$

where T_{rad} is the radiative subspace (perturbations that would generate radiative modes) and T_{silent} is the non-radiative subspace. Define the projection operators P_{rad}: $T_{\Phi s}Conf(\varphi) \to T_{rad}$ and P_{silent}: $T_{\Phi s}Conf(\varphi) \to T_{silent}$, satisfying $P_{rad} + P_{silent} = 1$ and $P_{rad} \cdot P_{silent} = 0$.

The projector P_{rad} is defined by the coupling functional: $P_{rad}\delta\varphi$ is the component of $\delta\varphi$ that couples to the radiative current $J_{rad}[\Phi_s]$. Since Φ_s is phase-silent (condition PS-3), $J_{rad}[\Phi_s] = 0$, and therefore the coupling map C: $T_{\Phi s}Conf \to Rad$ is identically zero. This implies $T_{rad} = \ker(C) = T_{\Phi s}Conf$ — but this cannot be right, since T_{rad} should be the subspace of radiative perturbations.

The resolution is that P_{rad} is defined not by the coupling to $J_{rad}[\Phi_s]$ (which is zero), but by the coupling to the radiative structure of the ambient admissibility geometry: $P_{rad}\delta\varphi$ is the component of $\delta\varphi$ that would be radiative in the absence of the phase-silent background, i.e., relative to the vacuum admissibility geometry. Formally:

$$\text{(A2)}\ P_{rad} = P_{rad}^{(vacuum)} \text{ restricted to the admissibility domain } Dom(A)|_{\Phi s}$$

The condition $M_{rad}(\Phi_s) = \varnothing$ then becomes: for all $\delta\varphi \in Dom(A)|_{\Phi s}$ (all admissibility-preserving perturbations of Φ_s), $P_{rad}\delta\varphi = 0$. Equivalently, $Dom(A)|_{\Phi s} \subset T_{silent}$. This means that the admissibility domain, when restricted to perturbations around Φ_s, lies entirely within the non-radiative subspace — no admissible perturbation of Φ_s can create a radiative mode.

The stability analysis of Theorem 54.2 follows from this mode decomposition. The second-order admissibility functional around Φ_s takes the form:

$$\text{(A3)}\ \delta^2 A[\Phi_s; \delta\varphi] = \int\int K_{AB}[\Phi_s](x,y)\ \delta\varphi_A(x)\ \delta\varphi_B(y)\ d^n x\ d^n y$$

where K_{AB} is the admissibility curvature kernel. For stability, we require $K_{AB}[\Phi_s] \leq 0$ (negative semi-definite) on T_{silent} — meaning all non-radiative perturbations return to the basin under the admissibility gradient force. This condition is satisfied for Φ_s in a deep admissibility basin, by the definition of the basin. On T_{rad}, the condition $P_{rad}\delta\varphi = 0$ for all admissible perturbations ensures that K_{AB} on T_{rad} is irrelevant: those directions are never accessed by admissible dynamics.

Appendix B: Comparison Table — Dark Matter Candidates

Candidate	Mechanism	Mass / Scale	Direct Detection	Indirect Detection	Collider Signature	Halo Profile	Small-Scale Structure	Status	Phase Theory Relation
WIMP	Thermal freeze-out	10 GeV – 10 TeV	Predicted; not seen ($\sigma < 10^{-47}$ cm^2)	Predicted; not seen ($\langle\sigma v\rangle < 10^{-26}$ cm^3/s)	MET +jets; not seen	NFW (cuspy)	Overpredicts satellites; cusp-core problem	Severely constrained; parameter space shrinking	Effective limit of radiatively active dark defects; not realized in Phase Theory
QCD Axion	PQ symmetry breaking; misalignment	10^{-6}–10^{-3} eV	Photon coupling only; ADMX limits	No annihilation; axion→photon in B field	None	CDM-like (cuspy)	Same as CDM	Open; some parameter space excluded	Shares photon coupling absent in phase-silent

Candidate	Mechanism	Mass / Scale	Direct Detection	Indirect Detection	Collider Signature	Halo Profile	Small-Scale Structure	Status	Phase Theory Relation
									structure
Fuzzy Dark Matter (ultra-light axion)	Misalignment; wave-like DM	~10^{-22} eV	None	None	None	Cored (quantum pressure)	Suppresses small-scale power; Lyman-α constraints	Marginally consistent; tension with Lyman-α	Cored halos similar to Phase Theory; different core mechanism
Sterile Neutrino	Oscillation production (Dodelson-Widrow)	keV range	None	X-ray decay line; not confirmed	None	Warm DM (cored)	Free-streaming suppresses small-scale power	3.5 keV line not confirmed; constrained	No decay channel in Phase Theory; distinct abundance mechanism
Primordial Black Holes (PBH)	Early-universe density perturbations	10^{-17}–10^{3} M_{sun}	Gravitational microlensing	Hawking radiation (small PBH)	None	Point-like; CDM halo	CDM-like	Various mass windows constrained; asteroid-	Not a phase-silent configuration; compact radia

Candidate	Mechanism	Mass / Scale	Direct Detection	Indirect Detection	Collider Signature	Halo Profile	Small-Scale Structure	Status	Phase Theory Relation
								mass window open	tive admissibility structures
SIDM (Self-Interacting DM)	Thermal freeze-out + self-interaction	GeV–TeV	Nuclear recoil (mediator-dependent)	Annihilation products	MET signatures	Cored (self-scattering)	Resolves core-cusp; some satellite issues remain	Active area; viable in some parameter space	Phase-silent self-interaction ($\sigma/m \sim 0.1\ cm^2/g$) analogous but topological in origin
MOND	Modified gravity ($a < a_0$)	N/A (no DM particle)	N/A	N/A	N/A	Flat rotation curves by construction	Fails for clusters without additional DM	Excellent for galaxies; fails for clusters and CMB	Emerges as approximate limit of Phase Theory in uniform Phase Genesis

Candidate	Mechanism	Mass / Scale	Direct Detection	Indirect Detection	Collider Signature	Halo Profile	Small-Scale Structure	Status	Phase Theory Relation
									environments
TeVeS	Relativistic MOND; scalar-vector-tensor	N/A + sterile ν	None (no DM particle)	None	None	MOND-like	Requires additional matter for CMB	Disfavored; GW speed constraint from GW170817	Effective field theory approximation to Phase Theory admissibility geometry
Phase-Silent Structure (Phase Theory)	Phase Genesis synchronization residues; topological freezing	Extended topological structure; $M_{adm} \sim 10^{7.5}$ M_{sun}	**σ = 0 (exact)**	**Zero (no annihilation channel)**	**None (categorical)**	Cored (admissibility pressure)	Suppressed below M_{adm}; resolves all three problems	All null results are confirmations; fully consistent with observations	Fundamental account; parent framework of all others as limiting cases

References

1. Zwicky, F. (1933). Die Rotverschiebung von extragalaktischen Nebeln. *Helvetica Physica Acta*, 6, 110–127.
2. Rubin, V. C., & Ford, W. K. (1970). Rotation of the Andromeda Nebula from a Spectroscopic Survey of Emission Regions. *Astrophysical Journal*, 159, 379–403.
3. van Albada, T. S., Bahcall, J. N., Begeman, K., & Sancisi, R. (1985). Distribution of dark matter in the spiral galaxy NGC 3198. *Astrophysical Journal*, 295, 305–313.
4. Bahcall, N. A., & Fan, X. (1998). The Most Massive Distant Clusters: Determining Ω and σ_8. *Astrophysical Journal*, 504, 1–6.
5. Kolb, E. W., & Turner, M. S. (1990). *The Early Universe*. Addison-Wesley, Redwood City, CA.
6. Peccei, R. D., & Quinn, H. R. (1977). CP Conservation in the Presence of Pseudoparticles. *Physical Review Letters*, 38(25), 1440–1443.
7. Weinberg, S. (1978). A New Light Boson? *Physical Review Letters*, 40(4), 223–226.
8. Wilczek, F. (1978). Problem of Strong P and T Invariance in the Presence of Instantons. *Physical Review Letters*, 40(5), 279–282.
9. Dodelson, S., & Widrow, L. M. (1994). Sterile neutrinos as dark matter. *Physical Review Letters*, 72(1), 17–20.
10. LUX Collaboration; Akerib, D. S., et al. (2017). Results from a search for dark matter in the complete LUX exposure. *Physical Review Letters*, 118, 021303.
11. PandaX-4T Collaboration; Meng, Y., et al. (2021). Dark Matter Search Results from the PandaX-4T Commissioning Run. *Physical Review Letters*, 127, 261802.

12. XENONnT Collaboration; Aprile, E., et al. (2022). First Dark Matter Search with XENONnT. *Physical Review Letters*, 131, 041003.
13. Fermi-LAT Collaboration; Ackermann, M., et al. (2015). Searching for Dark Matter Annihilation from Milky Way Dwarf Spheroidal Galaxies with Six Years of Fermi Large Area Telescope Data. *Physical Review Letters*, 115, 231301.
14. ATLAS Collaboration; Aad, G., et al. (2021). Search for new phenomena in events with an energetic jet and missing transverse momentum in pp collisions at $\sqrt{s}$ = 13 TeV with the ATLAS detector. *Physical Review D*, 103, 112006.
15. CMS Collaboration; Tumasyan, A., et al. (2021). Search for new particles in an extended dijet mass spectrum in pp collisions at $\sqrt{s}$ = 13 TeV. *Physical Review D*, 103, 092023.
16. Du, N., et al. (ADMX Collaboration) (2018). A Search for Invisible Axion Dark Matter with the Axion Dark Matter Experiment. *Physical Review Letters*, 120, 151301.
17. Boyarsky, A., Ruchayskiy, O., Iakubovskyi, D., & Franse, J. (2014). Unidentified Line in X-Ray Spectra of the Andromeda Galaxy and Perseus Galaxy Cluster. *Physical Review Letters*, 113, 251301.
18. Clowe, D., Bradač, M., Gonzalez, A. H., Markevitch, M., Randall, S. W., Jones, C., & Zaritsky, D. (2006). A Direct Empirical Proof of the Existence of Dark Matter. *Astrophysical Journal Letters*, 648, L109–L113.
19. Springel, V., et al. (2005). Simulations of the formation, evolution and clustering of galaxies and quasars. *Nature*, 435, 629–636.
20. Planck Collaboration; Aghanim, N., et al. (2020). Planck 2018 results. VI. Cosmological parameters. *Astronomy & Astrophysics*, 641, A6.

21. Mermin, N. D. (1979). The topological theory of defects in ordered media. *Reviews of Modern Physics*, 51(3), 591–648.
22. Tully, R. B., & Fisher, J. R. (1977). A new method of determining distances to galaxies. *Astronomy & Astrophysics*, 54, 661–673.
23. Hoekstra, H., & Jain, B. (2008). Weak Gravitational Lensing and Its Cosmological Applications. *Annual Review of Nuclear and Particle Science*, 58, 99–123.
24. Markevitch, M., et al. (2004). Direct Constraints on the Dark Matter Self-Interaction Cross Section from the Merging Galaxy Cluster 1E 0657-56. *Astrophysical Journal*, 606, 819–824.
25. Navarro, J. F., Frenk, C. S., & White, S. D. M. (1996). The Structure of Cold Dark Matter Halos. *Astrophysical Journal*, 462, 563–575.
26. de Blok, W. J. G. (2010). The Core-Cusp Problem. *Advances in Astronomy*, 2010, 789293.
27. Weinberg, D. H., Bullock, J. S., Governato, F., Kuzio de Naray, R., & Peter, A. H. G. (2015). Cold dark matter: Controversies on small scales. *Proceedings of the National Academy of Sciences*, 112(40), 12249–12255.
28. CDEX Collaboration; Liu, Z., et al. (2019). Constraints on Spin-Independent Nucleus Scattering with sub-GeV Weakly Interacting Massive Particle Dark Matter from the CDEX-1B Experiment at the China Jinping Underground Laboratory. *Physical Review Letters*, 123, 161301.
29. DarkSide-50 Collaboration; Agnes, P., et al. (2018). Low-Mass Dark Matter Search with the DarkSide-50 Experiment. *Physical Review Letters*, 121, 081307.

30. Hu, W., Barkana, R., & Gruzinov, A. (2000). Fuzzy Cold Dark Matter: The Wave Properties of Ultralight Particles. *Physical Review Letters*, 85, 1158–1161.

31. Milgrom, M. (1983). A modification of the Newtonian dynamics as a possible alternative to the hidden mass hypothesis. *Astrophysical Journal*, 270, 365–370.

32. McGaugh, S. S., Lelli, F., & Schombert, J. M. (2016). Radial Acceleration Relation in Rotationally Supported Galaxies. *Physical Review Letters*, 117, 201101.

33. Bekenstein, J. D. (2004). Relativistic gravitation theory for the modified Newtonian dynamics paradigm. *Physical Review D*, 70, 083509.

34. DESI Collaboration; Adame, A. G., et al. (2024). DESI 2024 VI: Cosmological Constraints from the Measurements of Baryon Acoustic Oscillations. arXiv:2404.03002.

35. Eisenstein, D. J., et al. (SDSS Collaboration) (2005). Detection of Baryon Acoustic Oscillations in the Redshift Survey of 46,748 Luminous Red Galaxies. *Astrophysical Journal*, 633, 560–574.

36. Beutler, F., et al. (2011). The 6dF Galaxy Survey: baryon acoustic oscillations and the local Hubble constant. *Monthly Notices of the Royal Astronomical Society*, 416, 3017–3032.

37. Bullock, J. S., Kolatt, T. S., Sigurdsson, S., Somerville, R. S., & Primack, J. R. (2000). Satellite Galaxies in the Milky Way Dark Matter Halo. *Astrophysical Journal*, 539, 517–521.

38. Boylan-Kolchin, M., Bullock, J. S., & Kaplinghat, M. (2011). Too big to fail? The puzzling darkness of massive Milky Way subhaloes. *Monthly Notices of the Royal Astronomical Society Letters*, 415, L40–L44.

39. Amaro-Seoane, P., et al. (LISA Collaboration) (2017). Laser Interferometer Space Antenna. arXiv:1702.00786.
40. Spergel, D. N., & Steinhardt, P. J. (2000). Observational Evidence for Self-Interacting Cold Dark Matter. *Physical Review Letters*, 84, 3760–3763.
41. Hanks, M. (2026). Effective Force Laws from Admissibility Gradients. *Phase Theory Series*, Paper 42. Dust LLC
42. Hanks, M. (2026). Gravitational Lensing from Phase-Topological Distortions. *Phase Theory Series*, Paper 45. Dust LLC
43. Hanks, M. (2026). Early-Universe Phase Synchronization Residues. *Phase Theory Series*, Paper 52. Dust LLC

Acknowledgments. The author thanks the foundations of topology and quantum field theory for providing the mathematical language adequate to the task of phase ontology. This work was completed June 3, 2026.

Correspondence: Marlon Hanks, Dust LLC, United States. Email:

Paper series note: This is Paper 54 of the Phase Theory series. Papers 42, 45, and 52 are cited explicitly in the text; the full series is available from the author. No external funding was received for this research.